

Bachelor Thesis IST-3



Stabilization of a rear-steered rider-controlled bicycle

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Abstract

Riding a bicycle with rear-wheel steering is almost impossible, even for experienced riders. The reason for the occurring difficulties will be investigated in this thesis. Based on this, it will be illustrated how such a bicycle has to be changed or how a rider has to behave to be able to stabilize it. Therefore an integrated approach to analyze the system will be considered, at which the human rider acts as a controller to keep the unstable bicycle with rear-wheel steering upright. Thus, purposeful methods of control theory will be applied to obtain wide-ranging statements about stability for solving the balancing problem, which will be visualized in numerous simulations.

Zusammenfassung

Ein Fahrrad mit Hinterradlenkung zu fahren ist selbst für geübte Fahrer fast unmöglich. Der Grund für die auftretenden Schwierigkeiten wird in dieser Arbeit näher beleuchtet. Darauf aufbauend wird dargestellt was konstruktiv geändert oder als Fahrer beachtet werden muss, um stabil fahren zu können. Im Zuge dessen wird ein ganzheitlicher Systemgedanke verfolgt, wobei der Mensch als Regler agiert um das instabile Fahrrad mit Hinterradlenkung aufrecht zu halten. Mit regelungstechnischen Methoden können somit umfassende Stabilitätsaussagen zur Lösung der Balancierproblematik getroffen werden, welche mit zahlreichen Simulationen veranschaulicht werden.

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List of Symbols and Abbreviations

Symbols

β	sideslip angle
δ	steer angle
$\tilde{\delta}$	input of the mechanical controller
δ_{rf}	steer angle of the rear-wheel plane when the steer axis is not vertical
λ	head angle
μ	phenomenological friction term between the controlled wheel and the ground
ν	course angle
ϕ	lean angle
ϕ_{m}	optimal phase margin
ϕ_{ref}	reference signal
ϕ_{rf}	lean angle of the rear-wheel plane when the steer axis is not vertical
ψ	yaw angle
ω	angular velocity
ω_{gc}	gain crossover frequency
ω_{w}	angular velocity of the controlled wheel
a	horizontal distance of the center of gravity (CG) of the system to the uncontrolled wheel
a_{b}	horizontal distance of the CG of the bicycle to the uncontrolled wheel
a_i	coefficients for the

	Hurwitz polynomial with $i = 1..n$
a_{r}	horizontal distance of the CG of the rider to the uncontrolled wheel
a_{s}	acceleration of the support line
a_{sr}	horizontal distance of the CG of the system to the uncontrolled wheel with a standing rider
c	spring constant
C_1, C_2	coefficients
d	damping coefficient
F_{cf}	centrifugal force
F_{k}	kink force
F_{r}	force induced by rider lean
F_{Y}	lateral force
$F_{\text{Y,W,front}}$	lateral force at the front-wheel in a wheel axis system
$F_{\text{Y,W,rear}}$	lateral force at the rear-wheel in a wheel axis system
F_{Z}	vertical force
$F_{\text{Z,W,rear}}$	vertical force at rear-wheel in a wheel axis system
g	gravitational acceleration
$G(s)$	plant
$G_{\phi\delta}(s)$	plant from δ to ϕ
$G_{\phi M}(s)$	plant from M to ϕ
$G_{\text{nmp}}(s)$	non-minimum phase part of a transfer function
h	height of the CG of the system in an upright position relative to the ground
h_{b}	height of the CG of the bicycle
h_{r}	height of the CG of the rider
h_{rb}	distance of the CG of the upper body to its axis of rotation
h_{sr}	h with a standing rider
J	moment of inertia of the system relative to the X -axis
J_{rb}	moment of inertia of the upper body of the rider relative to its axis of rotation
J_{rear}	moment of inertia of the rear-wheel and the rear fork relative to the steer axis

J_s	moment of inertia of the steering system relative to the steer axis
J_w	moment of inertia of the controlled wheel relative to its axis of rotation
k	spring constant
$k_\gamma, k_\delta, k_g, k_M, k_p$	proportional controllers
$K_m(s)$	transfer function of the mechanical controller
$K_r(s)$	transfer function of the rider
L	angular momentum
l	wheelbase
M	steer torque
m	total mass of the system
m_b	mass of the bicycle
m_r	mass of the rider
m_{rb}	mass of the upper body of the rider
m_{rear}	mass of the rear-wheel and the rear fork
M_{res}	resulting torque
m_s	mass of the steering system
m_w	mass of the controlled wheel
n_{gc}	slope at the gain crossover frequency
p_i	poles of a transfer function with $i = 1..n$
R	path radius
r_τ	castor offset
R_{cr}	radius of the chain rings of the steering mechanism
R_r	rocking radius of the tires
R_s	radius of a virtual cylinder containing the mass of the steering system
R_w	radius of the controlled wheel
T	time constant
t	time
v	forward velocity of the uncontrolled wheel
v_c	critical velocity for self-stabilization
X, Y, Z	intermediate axis system
X_E, Y_E, Z_E	earth-fixed axis system
z_i	zeros of a transfer function with $i = 1..n$

Abbreviations

CG	center of gravity
FSB	front-steered bicycle
FWS	front-wheel steering
IST	Institute for Systems Theory and Automatic Control
LHP	left-half-plane
RHP	right-half-plane
RSB	rear-steered bicycle
RWS	rear-wheel steering
SISO	single-input single-output

Chapter 1

Introduction and Outline

1.1 Introduction

Today, almost all bicycles have a front-wheel steering (FWS) but only few people wonder about this matter of fact. Thus, it seems to be an innovative approach to build a rear-steered bicycle (RSB), since new designs could be implemented like shorter chains for recumbent bikes [8, 31]. Additionally, it is safer to steer with the rear-wheel, as the major contribution of the braking force is on the front wheel [6].

A construction of a RSB can be found in an United States patent from 2001 [9]. The author promises superior handling and maneuverability of a RSB. But when trying to ride a bicycle with rear-wheel steering (RWS), significant difficulties can be experienced. The central problem is not what most people think, that a rider cannot get away from a pavement near him. Instead, a bicycle with RWS is extremely difficult to stabilize for a human rider because of a completely different dynamic behavior. The fact that a familiar system like a bicycle can show totally different riding characteristics is continuously fascinating people all over the world. Åström [5] describes this attribute as: "A design may be perfect statically but useless dynamically." Thus, if someone wanted to build an "unridable" bicycle similar to Jones [13], the solution would be a RSB.

The idea of RSBs exists since many decades. In 1870, Rankine [25] already analyzed a RSB and stated that there occurs a certain diffi-

culty of balancing it. A more empirical approach has been realized in 1977 by Laiterman [18] who built and tested a recumbent bike with a RWS and had severe problems riding it. Further analytical research including calculations of the eigenvalues of a rear-steered two-wheeler have been done in 1979 by Schwarz. He investigated a motorcycle with RWS for the United States National Highway Safety Administration in regard of safety improvements. Finally, he concluded that the instability was too fast to be stabilized by a human rider. However, the prototype of this motorcycle was safe in a different way, since no one could ride it [6, 21, 31]. After the turn of the millennium, with the papers of Åström et al. [6], Klein [15] and Lunze [7] typical methods of control theory have been applied on basic models of a RSB. With a mathematical description of the rider-bicycle-system, general statements about the reason of the difficulty of balancing a RSB and the ways to ameliorate the stability problems have been published. Åström investigated the dynamic restrictions and found that several parameters of the bicycle model have to be adapted [3]. According to Lunze [7], only in the recent investigations, a difference between RSBs and front-steered bicycles (FSBs) can mathematically be stated. The reason is that in earlier research the change of the steer angle at the handlebar has been assumed to be zero although it is the core of the completely different behavior of RSBs and FSBs.

In this bachelor's thesis models of Åström et al. [6], Lunze [7] and Zeller [32] will be taken into account to analyze and simulate the rider-controlled RSB with system-theoretical approaches. Besides, in several sections there will be a comparison to a FSB or a short explanation of self-stability to provide a better understanding of the different characteristics of a rider-controlled RSB and to investigate which relations can be applied on it.

Furthermore, the formulation of the balancing problem [6, 7] and the results of the limitations on non-minimum phase systems by Åström [3] will be used as a framework. Based on this, a new concept to stabilize a RSB by only using the steer angle will be extracted. With this concept, several theoretical and innovative techniques will be used in a new way, or will be developed to stabilize a RSB with simple control laws. Building on that, statements will be given what has to be changed on the bicycle and what has to be done by a rider to theoretically solve the balancing problem of a RSB.



Figure 1.1: Rear-steered bicycle of the Institute for Systems Theory and Automatic Control at the University of Stuttgart.

Finally, the purpose of this bachelor's thesis is as follows. On the one hand to find the reasons why a RSB is almost unridable. On the other hand, these reasons shall be used as a basis to develop several concepts to make the RSB of the Institute for Systems Theory and Automatic Control (IST) of Figure 1.1 theoretically rideable by a human rider. This implies that simple controllers have to be designed, that can be realized by a human.

1.2 Outline

The following thesis begins with an introduction of the central effects and characteristics of bicycle dynamics in Section 2.1. Afterwards, three models with increasing complexity will be derived in Section 2.2. They represent the basis for the following stability analyses. The

first one, called “Basic Model”, only shows the essential effects. This model will be investigated in Section 3.3 to find the reasons for the difficult stabilization and to find ways to stabilize a rider-controlled RSB. Amongst others, a possibility to stabilize a RSB Basic Model by steering and leaning will be explained. Afterwards, another way to balance a RSB Basic Model will be discussed, that takes the gyroscopic effect into account, which has been studied in detail by Klein and Sommerfeld in 1910 [14].

Then, the second model will be investigated in Section 3.4, which additionally considers forces on the static rear fork. It will be shown, that this model cannot be used in the case of a RWS. Therefore, the third model with a dynamic rear fork will be considered in Section 3.5, to find important effects that could have been neglected for the Basic Model. As calculations do not show additional useful regions in design space, the Basic Model will be regarded again for a controller design by only using the steer angle of the handlebar as rider input. For this design a basic principle will be applied that has been explored at analyses of the gyroscopic effect.

At first, a controller will be developed in Section 4.1, that describes how a rider should behave to be able to balance a RSB. Afterwards, a mechanical controller will be designed in Section 4.2 to support the human rider to make the stabilization easier and consequently realizable.

Chapter 2

Modeling of a Rear-Steered Bicycle

In the following chapter the most important effects and characteristics of bicycle dynamics for the balancing process will be introduced. Afterwards, three models of a bicycle with RWS with different complexity will be derived which will be the basis for the analyses of Chapter 3.

The first and simplest model is the basis for most investigations concerning stability and controller design for a rider-controlled RSB in this thesis. This model is based on a second order differential equation and approves a good insight into the most important forces and effects. Therefore, it provides an easy access to the fundamental problems that arise with a RWS. This model will be called the “Basic Model”.

To consider other neglected effects like self-stabilization, a further second order model with a static fork will be discussed. This so called “Static Fork Model” is based on the Basic Model. Analyses will show that some problems occur with this model for a configuration with RWS. Hence, the dynamics disregarded on the fork are covered by an additional fourth order model. This last model, which is based on the model by Zeller [32], will be called “Dynamic Fork Model”.

2.1 Effects and Characteristics

A FSB can be easily stabilized in many different lean angles. But, since balancing a RSB is much more difficult, stabilization in this thesis will stand for the return of the bicycle to the upright position of equilibrium with zero steer angle, after a disturbance.

Moreover, to be able to ride a bicycle with FWS the simple rule of thumb can be followed to steer in the direction of lean as stated in [17, 25], also called “lean-steer” [8]. This principle is based on the acceleration of the imaginary support line between the ground contact points of the wheels, back under the center of gravity (CG). This triggers several effects that can cause a stabilization of the bicycle. Furthermore, in the case of a riderless FSB certain geometric and dynamic characteristics are responsible for an “automatic” lean-steer behavior.

However, the behavior with a RWS is dramatically different compared to a FWS. For a rider-controlled and a riderless RSB central effects begin to work in different directions and therefore most bicycles with RWS are not rideable. Moreover, these and further effects and characteristics will be discussed in the following.

2.1.1 Centrifugal Force

In [17], it is stated that there are three central effects that relate to bicycle stabilization. The first effect is the acceleration of the support line caused by the steer angle. Basically, this leads to the centrifugal force, which acts on the CG of the moving system [6] and therefore causes a resulting torque. The centrifugal force is a so-called fictitious force as it has no counterforce [11]. Thus, it can only be noticed in the moving frame of reference where relative forces have to be considered. An observer in the inertial frame can only remark the centripetal forces at the wheels which keep the bicycle in a turn. The centrifugal force F_{cf} is given by

$$\vec{F}_{cf} = -m\vec{\omega} \times (\vec{\omega} \times \vec{r}). \quad (2.1)$$

Consequently, the centrifugal force accelerates the rotating object away from the center of rotation and therefore has the same direction for RSBs and FSBs. With the relation $\vec{v} = \vec{\omega} \times \vec{r}$ it can be concluded

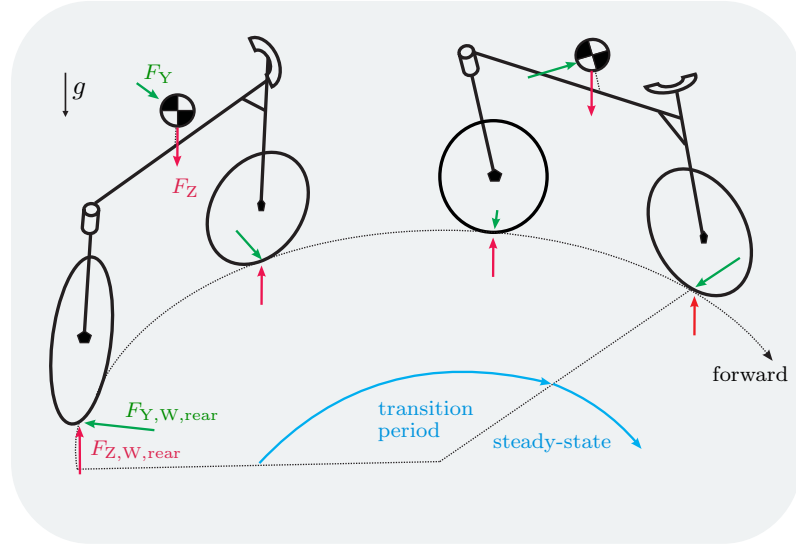


Figure 2.1: A RSB from transition period to steady-state: changing directions of the lateral tire force at the rear-wheel and of the lateral force on the CG.

that a rider is able to determine the centrifugal force by an adaption of the velocity or by adjusting the radius r of the turn by steering. Moreover, an appropriate variation of the centrifugal force is very important for stabilization.

2.1.2 Kink Force

The second dominant term according to [17] is caused by the acceleration of the support line due to steer rate. This effect basically traces back to the lateral force $F_{Y,W}$ that occurs at the steered wheel at a change of the steer angle (compare Figure 2.1 and [18]). This fictitious force [32] also causes a resulting torque on the CG of the moving system [6] and helps to stabilize a FSB by applying an appropriate lean-steer [28]. This force will be called “kink force” to simplify the following explanations [32].

If one wanted to recover from a tilt of a RSB to the left by bringing the support line back under the CG of the system, he should initially steer to the right to move the rear-wheel to the left [8, 25, 30]. This would induce a destabilizing centrifugal force which causes a fall to the left.

In opposition, if the rider would steer into the lean this would initially increase the lean angle in contrast to a FSB [18]. The reason is that the rear-wheel of a RSB has to turn right to steer the bicycle to the left. Therefore, the lateral tire force $F_{Y,W}$ at the steered wheel and with this the kink force on the CG will act in the opposite direction for a RSB. And as the kink force, which is proportional to the steer rate, increases faster than the centrifugal force, which is related to the steer angle, the lean angle will increase in the beginning until the centrifugal force gets big enough to compensate this effect. Since this is generally too late, a stabilization of a RSB is very difficult. Finally, these analyses showed that the change in direction of the kink force causes inherent problems to ride a RSB and a rule of thumb like simply steering into a lean as for a FSB cannot be stated for a RSB.

The circumstance that the kink force and the centrifugal force work against each other with a changing predominance can be described as an “inverse response behavior”. This can be seen in Figure 2.1, at which the lateral force F_Y on the CG changes its direction relatively to the frame.

Since the kink force acts only at a change of the steer angle, which is called “transition period”, it is zero when the steer angle is constant and the stabilized bicycle is in a “steady-state”.

Besides, the kink force is dominant when the velocity is small and enables a rider on a FSB to balance even with small velocities. The effect of the kink force can be increased by leaning forward. It is therefore easier to balance a FSB while driving steep hills when moving the body forward to the handlebar [32].

Finally, the acceleration of the support line a_s caused by the steer rate $\dot{\delta}$ and the steer angle δ of the rear-wheel, is summarized in Table 2.1 for different types of bicycles with the coefficients C_1 and C_2 [17]. As a result, it is easier to ride a bicycle with both-wheels steering than with RWS, because there are no complications among steer rate and angle. Additionally, in the case of a FSB, counter steering is very important for obstacle avoidance since the kink force and the

Table 2.1: Acceleration of the support line.

type	acceleration	description
front-wheel steering	$a_s = C_1\delta + C_2\dot{\delta}$	effects rectified
rear-wheel steering	$a_s = C_1\delta - C_2\dot{\delta}$	effects opposed
both-wheels steering	$a_s = C_1\delta$	only steer angle

centrifugal force act in the same direction. Thus, if a rider wants to turn to the left he should first counter steer to the right. Therefore, both forces will tilt the bicycle to the left. That way, a much sharper turn can be realized in contrast to only using rider lean. In the case of motorcycles, counter steering is often essential since the effect of rider lean is not as large as it is for a much lighter bicycle [5, 6].

2.1.3 Gravity Torque

The third and last dominant term illustrated in [17] is the gravity torque. In general, this torque is destabilizing the system. Nevertheless, clever distribution of mass can also provide self-stability, as shown in [16]. This article describes a two-mass front-steered skate at which a lean-steer behavior is realized only through the mass of the steering system that is much lower in contrast to the frame. Thus, it falls faster than the frame causing the front-wheel to incline in the lean and consequently a self-stable configuration can be obtained.

Besides, the gravity torque can be actively used by a rider who leans relatively to the frame of the bicycle. This additional input can help to stabilize a RSB [30], which will be illustrated in Section 3.3.3.

2.1.4 Gyroscopic Effect

Although the gyroscopic effect and the castor offset of Section 2.1.5 can provide self-stabilization by lean-steer which is also induced by the rider, it has been approved that rider-controlled stability is related to self-stability of an uncontrolled bicycle [16]. The reason is that dis-

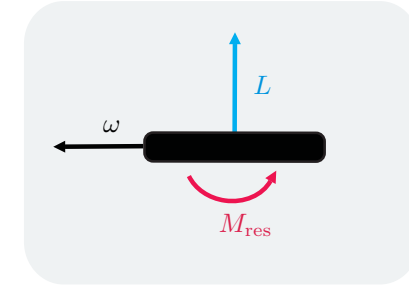


Figure 2.2: Resulting moment induced by a tilt on a spinning wheel.

turbances are reduced automatically and consequently the following consideration will be useful even for a rider-controlled bicycle.

The famous gyroscopic effect of spinning wheels has often been stated to be necessary for self-stability of a bicycle with FWS, since Klein and Sommerfeld concluded this from their equations in [14]. They stated that this effect causes a lean-steer and therefore triggers the centrifugal force which counteracts a fall.

If one considers a wheel that rolls forward with the angular momentum L and tries to tilt it to the left with the angular velocity ω , it will turn to the left. The reason for this effect is that tilting a spinning wheel actually corresponds to a change of the direction of the angular momentum, which causes a resulting torque M_{res} . This relation is given by

$$\frac{d\vec{L}}{dt} = \vec{M}_{\text{res}}. \quad (2.2)$$

The gyroscopic effect is indeed the reason for self-stabilization of a single rolling wheel. However, there have been recent research results [17, 21] that the gyroscopic effect can be necessary for self-stability but not in general, what is stated misleadingly in [14].

Moreover, the gyroscopic effect of the relatively lightweight wheels is usually not central for a rider-controlled bicycle, since it is small compared to the three effects presented in the sections before [7, 32]. Nevertheless, in Chapter 3, it will be shown that a very useful principle can be derived from the gyroscopic effect, that helps to

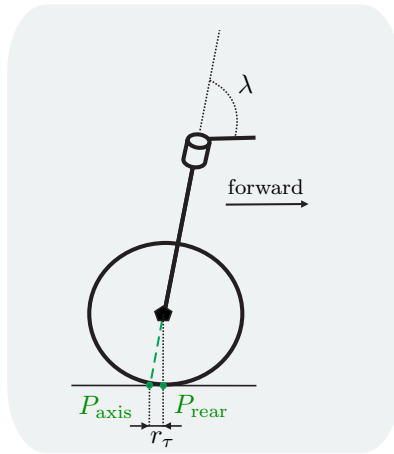


Figure 2.3: Negative castor offset at the rear-wheel of a RSB.

stabilize a RSB.

Besides, it is important to keep in mind that the gyroscopic effect tends to steer right when leaning to the left for a RSB in contrast to a FSB.

2.1.5 Castor Offset

In his famous work [13] Jones finds the **castor offset to be very important for bicycle stability**, because he could not manage riding a FSB hands-off with a negative castor offset.

Such a negative castor offset r_τ is shown in Figure 2.3, where λ denotes the head angle. The castor offset r_τ is the distance between the ground contact point of the rear-wheel P_{rear} and the intersection point P_{axis} of the virtual extension of the steer axis of an upright bicycle with zero steer angle and the ground. If P_{axis} is behind of P_{rear} , we will call this a “negative” castor offset. To find out if the castor offset is positive or negative one can simply hold a bicycle at the saddle and lean it to a side [6]. The direction in which the steered wheel inclines shows the “direction” of the castor offset.

Moreover, in [13] two effects of the positive castor offset for a FSB

have been discovered. On the one hand, a self-centering of the controlled wheel can be realized, similar to a shopping cart wheel. On the other hand, the normal force at the ground induces a feedback from the lean to the steer angle, that can stabilize the bicycle by realizing a lean-steer behavior [6, 18]. This will be illustrated in detail in Section 3.4.

This lean-steer characteristics can also be obtained in terms of an autonomous RSB with a negative castor offset. The centripetal forces on the wheels in the steady-state shown in Figure 2.1, act in lateral direction to the center of rotation of the bicycle and cause a self-centering behavior.

Before this steady-state can actually be achieved there are fundamental differences in the transition period between RWS and FWS. The reason is the opposite direction of the lateral tire force at a change of the steer angle with a RWS, as stated in Section 2.1.2.

When a RSB with a negative castor offset begins to lean to a side, the normal force on the rear-wheel will cause a lean-steer. Then, the lateral tire force at the rear-wheel destabilizes it, due to an additional steering into the lean. In a first scenario, the steer angle and the centrifugal force will be sufficiently large to tip the bicycle over in the opposite direction. In a second scenario, at which the RSB is slow, it tips over in the direction in which the tilt was initiated. This is due to the fact, that the centrifugal force is proportional to the velocity to the power of two.

Both scenarios can be easily tested by pushing a FSB backwards. Thus, a steady-state cannot be reached with an autonomous RSB. For a rider-controlled RSB with negative castor offset the impact of the forces are the same as for the autonomous RSB, see Table 2.2.

If we change the castor offset on the RWS to a positive configuration, we obtain a stabilizing steer torque for a rider-controlled RSB in the transition period [30].

In contrast to this, we get a reverse lean-steer for an autonomous RSB [8]. As a result, the centrifugal force will knock the bicycle over in the direction into which a fall is initiated before a steady-state can be reached. Moreover, a positive castor offset tends to destabilize the RSB in the steady-state for a controlled system, since the centripetal

Table 2.2: Effect of castor offset on a rider-controlled RSB.

type of castor offset	lateral tire force	
	in transition	in steady-state
positive castor offset reverse lean-steer	decrease of steer angle	increase of steer angle
negative castor offset lean-steer	increase of steer angle	decrease of steer angle
zero castor offset no lean-steer	no effect on steering	no effect on steering

force at the steered wheel increases the steer angle.

Finally, there is no optimal set up of the castor offset for a RSB, since

“..transient and steady-state lateral tire forces are in opposite directions, and thus apply steer torques of opposite sign with any caster offset.” [30]

This is illustrated in Table 2.2. Therefore, a method recommended in [18] is to minimize the castor offset for a RSB to eliminate its effects. This concept will be used for the Basic Model in Section 2.2.1.

2.2 Inverted Pendulum Models

To be able to analyze the difficulty of riding a RSB, the appropriate models for the balancing process have to be derived. The models discussed in this thesis and the stabilization process of a bicycle have several characteristics in common with the challenge of keeping an inverted pendulum upright. Moreover, the derivation will show that the Basic Model will even have the same capsized mode as an inverted pendulum [19]. Therefore, the models will be called “inverted pendulum models”.

2.2.1 Basic Model

Assumptions

The balancing and steering of bicycles is a very complex process. To keep focused on the main issues for control investigations, the Basic Models used in [6, 7, 28, 32] will be considered. To reproduce the bicycle for the purpose of this work several neglects and assumptions, some already indicated in Section 2.1, have to be taken into account. Amongst others, the following central assumptions hold for the Basic Model:

- The bicycle consists of four rigid parts: two wheels, a frame and a rear fork.
- The mass of the system is modeled as a point mass m .
- The tires do not slide, the wheels always have contact to the ground and the process is frictionless.
- The wheels are infinitely slim. An exception will be considered in Section 3.3.2.
- The bicycle is driven on a horizontal plane.
- The rider does not move his upper part of the body with respect to the frame and is modeled as a rigid body. Consequently, rider lean is no control variable. However, a detailed analysis of rider lean can be found in Section 3.3.3.
- The forward velocity of the uncontrolled wheel is constant.
- **The only control variable is the steer angle δ .** However, the steer torque M will be used instead of the steer angle δ in Section 3.4 and 3.5.
- The only degree of freedom of the system is the lean angle ϕ , which leads to a single-input single-output (SISO) system.
- The angles δ and ϕ are small, so that the equations can be linearized.

- The rear-wheel turns right when steering to the left. And the steer angle is transferred 1:1 from the handlebar to the rear fork. A more complex transmission will be considered in Section 4.2.
- The gyroscopic effect of the spinning wheels, described in Section 2.1.4, is neglected since it is generally small. Nevertheless, a more detailed investigation will be presented in Section 3.3.4.
- The castor offset (compare Section 2.1.5) and accordingly the distance between P_{rear} and P_{axis} in Figure 2.3 is set to zero. Therefore, the impact of the tire forces on the steer angle can be minimized to be able to focus on the essential effects.
- The head angle λ of the rear fork is 90° , which implies that the steer axis is vertical.

Geometry and Dynamics

The derivation of the Basic Model of the RSB is mainly based on [7] and [6]. Similar figures and models can also be found in [19] and [28].

Figure 2.4 shows the geometry of a RSB, which follows the ISO 8855 standard [1]. The axes XYZ represent an intermediate axis system and the axes $X_E Y_E Z_E$ an earth-fixed axis system. Both are right-handed Cartesian coordinate systems. The intermediate axis system is fixed to the ground contact point P_{front} of the front-wheel. The Z -axis points in the Z_E -direction, the XY -plane is located in the $X_E Y_E$ -plane and the X -axis goes through the ground contact points of both wheels.

The horizontal distance of the CG to P_{front} is denoted by a , the height of the CG in an upright position relative to the ground is denoted by h and the wheelbase will be l . The lean angle ϕ between the vertical plane and the plane of the frame is positive around the positive X -axis when leaning to the right. The steer angle δ between the plane of the rear-wheel and the plane of the frame is positive around the negative Z -axis when steering to the left.

In the following, all angles are assumed to be positive. In Figure 2.5, ψ denotes the angle between the X - and the X_E -axis, just like between the Y - and the Y_E -axis. The angles β and ν describe the

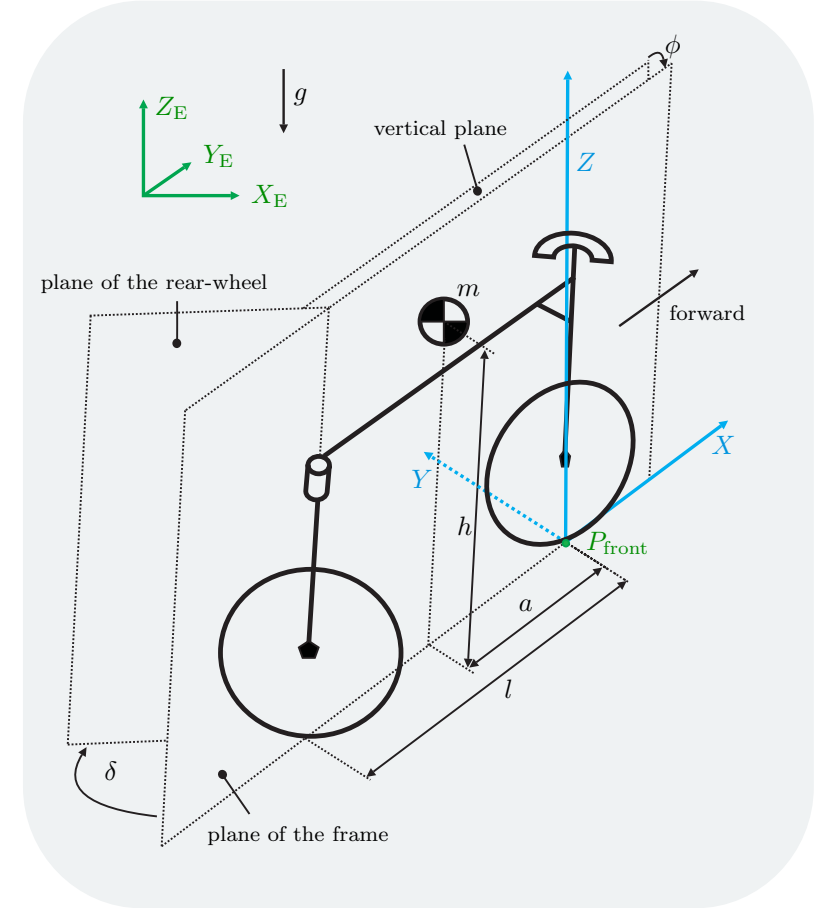


Figure 2.4: Geometry of the Basic Model of a RSB.

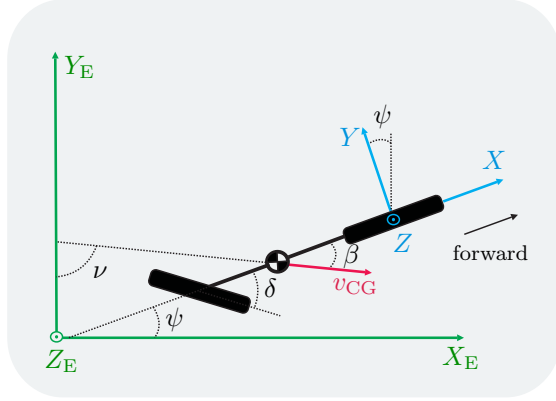


Figure 2.5: Kinematics of a RSB relative to the intermediate and the earth-fixed axis system.

direction of the velocity v_{CG} of the CG relative to the intermediate and the earth-fixed axis system respectively. The relation between these angles is given by

$$\beta - 90^\circ = \psi - \nu. \quad (2.3)$$

As illustrated in Figure 2.6, the virtual extension of the axes of rotation of the wheels intersect in the instantaneous center of rotation P_{icr} on the Y -axis. Thus, δ determines P_{icr} around which the bicycle is turning.

The velocity v_{CG} is perpendicular to the connection R between P_{icr} and the CG. Further geometric relations that follow from Figure 2.4 and 2.6 can be expressed as

$$a = R \sin \beta \quad (2.4)$$

and

$$\tan \beta = \frac{a}{l} \tan \delta. \quad (2.5)$$

The vector $\vec{v}_{CG}$ of the velocity of the CG can be decomposed into v in X -direction and v_Y in Y -direction. Here, v is the constant velocity

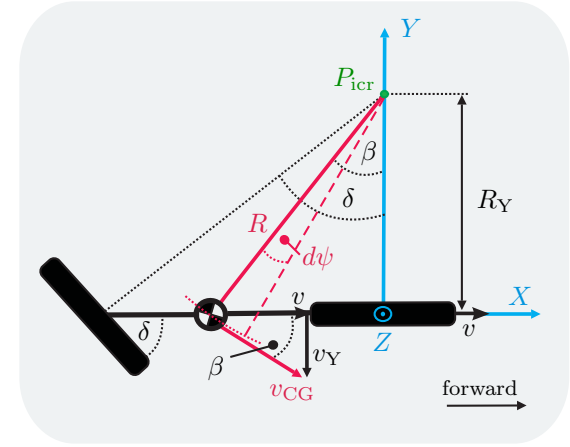


Figure 2.6: Kinematics of a RSB relative to the instantaneous center of rotation.

of the front-wheel in X -direction and v_Y is caused by a lean of the RSB. With the kinematic relations of Figure 2.6

$$v_{CG} = \frac{v}{\cos \beta} \quad (2.6)$$

can be concluded.

Due to the turning of the RSB around P_{icr} , the intermediate axis system changes its position relative to the earth-fixed system by the infinitesimal small angle $d\psi$ in the time dt . This can be seen in Figure 2.6, at which the motion of the bicycle leads to a change of the direction of R indicated by the red dashed line. Using a simple Taylor linearization for $\sin d\psi$ at $d\psi = 0$

$$\sin d\psi = \dot{\psi} = \frac{v_{CG}}{R} \xrightarrow{(2.6)} \dot{\psi} = \frac{v}{R \cos \beta} \quad (2.7)$$

can be found.

The aim for the balancing process is to state an equation that only depends on the input δ and the output ϕ . Furthermore, the acceleration in the Y -direction is needed, to be able to set up a differential

equation based on forces and torques. Therefore, the vector $\dot{\vec{v}}_{CG}$ has to be projected on the unit vector $\vec{Y}_u$ in Y -direction. Written in the earth-fixed axis system

$$\vec{v}_{CG} = v_{CG} \begin{pmatrix} \sin \nu \\ -\cos \nu \end{pmatrix} \quad \text{and} \quad \vec{Y}_u = \begin{pmatrix} -\sin \psi \\ \cos \psi \end{pmatrix}$$

follows from Figure 2.5 and with the product rule

$$\dot{\vec{v}}_{CG} = \dot{v}_{CG} \begin{pmatrix} \sin \nu \\ -\cos \nu \end{pmatrix} + v_{CG} \dot{\nu} \begin{pmatrix} \cos \nu \\ \sin \nu \end{pmatrix} \quad (2.8)$$

can be found. Finally, the projection is given by

$$\begin{aligned} \dot{\vec{v}}_{CG} \vec{Y}_u &= -\dot{v}_{CG} \sin \nu \sin \psi - \dot{v}_{CG} \cos \nu \cos \psi \\ &\quad - v_{CG} \dot{\nu} \cos \nu \sin \psi + v_{CG} \dot{\nu} \sin \nu \cos \psi. \end{aligned} \quad (2.9)$$

Using the addition theorems for trigonometric functions and Equation (2.3)

$$\dot{\vec{v}}_{CG} \vec{Y}_u = -\dot{v}_{CG} \cos(-\nu + \psi) - v_{CG} \dot{\nu} \sin(-\nu + \psi) \quad (2.10)$$

$$= -\dot{v}_{CG} \sin \beta + v_{CG} \dot{\nu} \cos \beta \quad (2.11)$$

$$= -\dot{v}_{CG} \sin \beta + v_{CG} \dot{\psi} \cos \beta - v_{CG} \dot{\beta} \cos \beta \quad (2.12)$$

can be gained. With the derivative of Equation (2.6), which is

$$\dot{v}_{CG} = \frac{v \sin \beta}{\cos^2 \beta} \dot{\beta}, \quad (2.13)$$

$\dot{v}_{CG}$ can be substituted in Equation (2.12) and with Equation (2.7) this yields

$$\dot{\vec{v}}_{CG} \vec{Y}_u = -v \left(\frac{\sin^2 \beta}{\cos^2 \beta} + 1 \right) \dot{\beta} + v_{CG} \dot{\psi} \cos \beta \quad (2.14)$$

$$= \frac{-v}{\cos^2 \beta} \dot{\beta} + \frac{v_{CG} v}{R}. \quad (2.15)$$

Using the Equations (2.4), (2.5) and (2.6) we obtain

$$\dot{\vec{v}}_{CG} \vec{Y}_u = \frac{-v}{\cos^2 \beta} \dot{\beta} + \frac{v^2}{l} \tan \delta. \quad (2.16)$$

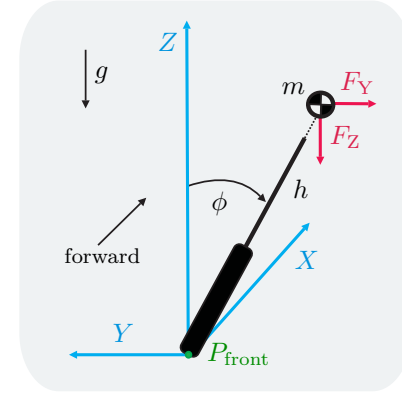


Figure 2.7: Inverted pendulum: rear view of a leaning RSB.

The derivative of (2.5), that is

$$\frac{1}{\cos^2 \beta} \dot{\beta} = \frac{a}{l} \frac{1}{\cos^2 \delta} \dot{\delta}, \quad (2.17)$$

finally leads to

$$\dot{\vec{v}}_{CG} \vec{Y}_u = -\frac{a v}{l} \frac{1}{\cos^2 \delta} \dot{\delta} + \frac{v^2}{l} \tan \delta, \quad (2.18)$$

which is only a function of the input δ .

To describe the balancing process of the RSB, the YZ -plane has to be considered. The rear-view in Figure 2.7 shows a rotation of the RSB around the positive X -axis which leads to a positive lean angle ϕ . With the forces F_Y , F_Z and the height h , the principle of angular momentum in X -direction around P_{front} can be stated as

$$J \ddot{\phi} = F_Z h \sin \phi + F_Y h \cos \phi, \quad (2.19)$$

at which J denotes the moment of inertia of the system and $F_Y = m \dot{\vec{v}}_{CG} \vec{Y}_u$ holds for the force in Y -direction.

Moreover, the forces F_Z and F_Y are

$$F_Z = mg, \quad (2.20)$$

where g is the gravitational acceleration and

$$F_Y = m \ddot{\vec{v}}_{CG} \vec{Y}_u \xrightarrow{(2.18)} F_Y = -\frac{mav}{l} \frac{1}{\cos^2 \delta} \dot{\delta} + \frac{mv^2}{l} \tan \delta. \quad (2.21)$$

In Figure 2.7, the direction of F_Y is defined by the centrifugal force, which acts in the negative Y -direction if a positive steer angle δ to the left is applied on the handlebar. To prove this, one can consider the acceleration in Y -direction of Equation (2.18), at which only $\frac{v^2}{l} \tan \delta$ is dependent on δ after a linearization around the zero position. Thus, this term has to be proportional to the centrifugal force, since this force is a function of the radius of the turning circle and therefore as well of δ . Given that the term in Equation (2.18), which is proportional to δ , is positive and assuming $\dot{\delta}$ to be zero, the direction of the arrow of F_Y in Figure 2.7 has to point in the direction of the centrifugal force, which is away from the center of rotation in the negative Y -direction. If $\dot{\delta}$ is not zero, the direction of F_Y would be the same.

With the parallel axes theorem the moment of inertia J of the point mass relative to the X -axis can be approximated as $J \approx mh^2$. Inserting the forces and J in Equation (2.19), the differential equation

$$mh^2 \ddot{\phi} - mgh \sin \phi = \left(-\frac{mav}{l} \frac{1}{\cos^2 \delta} \dot{\delta} + \frac{mv^2}{l} \tan \delta \right) h \cos \phi \quad (2.22)$$

for the balancing process of a RSB can be obtained. For small angles a Taylor linearization can be applied at the upright position $\phi_0 = 0$ on a straight ahead path $\delta_0 = 0$, which is the desired steady-state for the stabilization in this thesis. The final equation of motion of the RSB is given by

$$\ddot{\phi} - \frac{g}{h} \phi = -\frac{av}{hl} \dot{\delta} + \frac{v^2}{hl} \delta. \quad (2.23)$$

In fact, this second order model is only dependent on the lean angle ϕ and the steer angle δ . Here, the second term describes the influence of the gravity torque, the third term describes the influence of the kink force and the last term depends on the centrifugal force [5, 7].

Another way to get to this equation is illustrated in [6]. Here, the angular momentum is derived to set up the principle of angular

momentum, with the gravity and centrifugal force causing external torques.

Besides, to be able to compare the behavior of a RSB to a FSB, the differential equation of a bicycle with FWS is necessary. Since a RSB is a FSB with a negative velocity, only v at the uncontrolled wheel in Equation (2.23) changes its sign [6]. This gives

$$\ddot{\phi} - \frac{g}{h} \phi = \frac{av}{hl} \dot{\delta} + \frac{v^2}{hl} \delta. \quad (2.24)$$

It can easily be observed, that the kink force changes its direction as described in Section 2.1.2. Further analyses in Chapter 3 will point out the problematic implications of this different direction of the kink force for a RSB compared to a FSB.

2.2.2 Static Fork Model

To be able to model a more complex dynamic behavior like self-stabilization, a model that contains additional details is necessary. In [6], the Static Fork Model uses the Basic Model of Section 2.2.1, with a more detailed rear fork. Therefore, the same assumptions as listed in Section 2.2.1 will be taken into account, but with a change of the simplifications for the **castor offset**, the **head angle** and the **steer torque**.

In Figure 2.8, $r_\tau \neq 0$ denotes the castor offset and $\lambda \neq 90^\circ$ denotes the head angle. The X -axis is in the plane of the frame and goes through P_{front} and P_{axis} . Furthermore, instead of the steer angle δ the rider input will now be the **steer torque M** applied on the handlebar.

According to the name of the model, the rear fork is in a static torque balance. This central simplification is justified by the faster dynamics of the fork compared to the frame [5]. Thus, the rear fork is considered to be always in a steady-state. Consequently, the lateral tire force $F_{Y,W,\text{rear}}$ points to the center of rotation of the bicycle (compare Figure 2.1). This lateral force $F_{Y,W,\text{rear}}$ and the normal force $F_{Z,W,\text{rear}}$ at the tire are under certain conditions able to cause a lean-steer behavior and provide self-stability of a FSB. The topic of self-stability will be illustrated in detail in Section 3.4.1.

On the one hand, the steer angle δ changes to δ_{ff} . On the other hand, an additional torque around the X -axis is induced by the normal force $F_{Z,W,\text{rear}}$ and the castor offset r_τ , which can be seen in Figure 2.9. After a linearization, this torque can be expressed as

$$-F_{Z,W,\text{rear}}r_\tau\delta_{\text{ff}} = -F_{Z,W,\text{rear}}r_\tau \sin \lambda \delta = -\frac{mgar_\tau \sin \lambda}{l}\delta. \quad (2.37)$$

Finally, this leads to the modified Basic Model for the frame that is

$$\ddot{\phi} - \frac{g}{h}\phi = -\frac{av \sin \lambda}{hl}\dot{\delta} + \frac{(v^2h - ar_\tau g) \sin \lambda}{h^2l}\delta. \quad (2.38)$$

A combination of Equation (2.36) of the fork and Equation (2.38) of the frame yields

$$\begin{aligned} & \ddot{\phi} - \frac{avg \sin \lambda}{(v^2 \sin \lambda - lg \cos \lambda)h}\dot{\phi} + \frac{g^2(hl \cos \lambda - ar_\tau \sin \lambda)}{(v^2 \sin \lambda - lg \cos \lambda)h^2}\phi \\ &= -\frac{alv}{(v^2 \sin \lambda - lg \cos \lambda)mahr_\tau}\dot{M} + \frac{l(v^2h - ar_\tau g)}{(v^2 \sin \lambda - lg \cos \lambda)mah^2r_\tau}M, \end{aligned} \quad (2.39)$$

which is the final equation for the balancing process of the Static Fork Model.

To be able to compare a RSB to a FSB, the Static Fork Model for a bicycle with FWS is needed. Analog to Equation (2.24), the equation for a FSB is given by a change of the direction of the velocity v , which leads to

$$\ddot{\phi} - \frac{g}{h}\phi = \frac{av \sin \lambda}{hl}\dot{\delta} + \frac{(v^2h - ar_\tau g) \sin \lambda}{h^2l}\delta \quad (2.40)$$

for the frame.

The equation of the fork stays the same since all velocities are quadratic in Equation (2.36). Thus, the combination of the frame and the static front fork is

$$\begin{aligned} & \ddot{\phi} + \frac{avg \sin \lambda}{(v^2 \sin \lambda - lg \cos \lambda)h}\dot{\phi} + \frac{g^2(hl \cos \lambda - ar_\tau \sin \lambda)}{(v^2 \sin \lambda - lg \cos \lambda)h^2}\phi \\ &= \frac{alv}{(v^2 \sin \lambda - lg \cos \lambda)mahr_\tau}\dot{M} + \frac{l(v^2h - ar_\tau g)}{(v^2 \sin \lambda - lg \cos \lambda)mah^2r_\tau}M. \end{aligned} \quad (2.41)$$

In Chapter 3, the Static Fork Model will be investigated in detail and several characteristics and problems of this model for a RSB and a FSB will be illustrated.

2.2.3 Dynamic Fork Model

In the following, a third model will be introduced given by Zeller [32], who refers to [14, 25, 29] and [13]. This fourth order model is based on the Static Fork Model. Nevertheless, there are several significant differences for the fork.

Amongst others, the rear fork shows a dynamic behavior to avoid the weaknesses of a simplified static fork. Additionally, the rear fork is not massless anymore and the gyroscopic effect caused by the lean rate $\dot{\phi}$ and the friction at the ground will be taken into account. All other assumptions are similar to the Static Fork Model.

However, the Dynamic Fork Model will not be derived in this context since it is not central for the following chapters. Instead, the sign of the velocity v in the equation of the dynamic front fork from [32] will be changed analog to the Basic and Static Fork Model, to gain a model of the dynamic rear fork. Thus, the equation of the dynamic rear fork is expressed by

$$\begin{aligned} J_s \ddot{\delta} + \mu \dot{\delta} + \left(-\frac{mgar_\tau \cos \lambda}{l} + \frac{mKav^2r_\tau \sin \lambda}{l^2} \right) \delta \\ = \frac{J_w v \sin \lambda}{R_w} \dot{\phi} - \frac{mgar_\tau \sin \lambda}{l} \phi. \end{aligned} \quad (2.42)$$

According to [32], the first term of Equation (2.42) is induced by a change of the angular momentum of the steering system. The second term is a friction term basically caused by the ground contact of the rear tire and both terms which are proportional to g are obviously induced by the gravity acting on the mass of the system. Furthermore, the term proportional to v^2 is the torque caused by the centrifugal force and the term proportional to $\dot{\phi}$ is caused by the gyroscopic effect.

Moreover, J_s is the moment of inertia of the steering system relative to the steer axis and includes the rear-wheel, rear fork and the handlebar. It will be approximated as $J_s \approx \frac{1}{4}m_s R_s^2$, at which m_s is

the mass of the steering system and R_s the radius of a virtual cylinder containing this mass. Besides, J_w is the moment of inertia of the rear-wheel relative to its axis of rotation. A similar approximation leads to $J_w \approx \frac{1}{2}m_w R_w^2$, at which m_w is the mass and R_w the radius of the rear-wheel. Additionally, the factor μ is a friction coefficient and the factor K is given as

$$K = 1 + \frac{lm_w R_w \cos \lambda}{2ar_\tau m}. \quad (2.43)$$

Inserting these relations in Equation (2.42) yields

$$\begin{aligned} & \frac{1}{4}m_s R_s^2 \ddot{\delta} + \mu \dot{\delta} \\ & + \left(-\frac{mgar_\tau \cos \lambda}{l} + \frac{mav^2 r_\tau \sin \lambda}{l^2} \left(1 + \frac{lm_w R_w \cos \lambda}{2ar_\tau m} \right) \right) \delta \quad (2.44) \\ & = \frac{m_w R_w v \sin \lambda}{2} \dot{\phi} - \frac{mgar_\tau \sin \lambda}{l} \phi \end{aligned}$$

for the dynamic rear fork.

The equation for the frame of the Dynamic Fork Model of a RSB is identical to the frame of the Static Fork Model, which is

$$\ddot{\phi} - \frac{g}{h} \phi = -\frac{av \sin \lambda}{hl} \dot{\delta} + \frac{(v^2 h - ar_\tau g) \sin \lambda}{h^2 l} \delta. \quad (2.45)$$

Moreover, in the case of a FSB

$$\ddot{\phi} - \frac{g}{h} \phi = \frac{av \sin \lambda}{hl} \dot{\delta} + \frac{(v^2 h - ar_\tau g) \sin \lambda}{h^2 l} \delta \quad (2.46)$$

is the equation of the frame similar to the Static Fork Model and

$$\begin{aligned} & \frac{1}{4}m_s R_s^2 \ddot{\delta} + \mu \dot{\delta} \\ & + \left(-\frac{mgar_\tau \cos \lambda}{l} + \frac{mav^2 r_\tau \sin \lambda}{l^2} \left(1 + \frac{lm_w R_w \cos \lambda}{2ar_\tau m} \right) \right) \delta \quad (2.47) \\ & = -\frac{m_w R_w v \sin \lambda}{2} \dot{\phi} - \frac{mgar_\tau \sin \lambda}{l} \phi \end{aligned}$$

can be stated for the dynamic front fork, which corresponds to [32].

Chapter 3

Analysis of a Rear-Steered Bicycle

In the following, the models derived in Chapter 2 will be analyzed and simulated with system theoretic concepts and methods. With these approaches, the reason for the difficult stabilization of a RSB will be investigated. Additionally, several ways to make the stabilization easier and several ways to theoretically stabilize a RSB will be illustrated.

Besides, the goal of these techniques is to provide control laws that are as easy as possible to be feasible by a human rider. Furthermore, some analyses and simulations will be compared to FSBs to show the dramatic difference between RWS and FWS.

For the simulations and explanations in the following chapters it is useful to keep the positive directions of the steer angle δ , the lean angle ϕ and the steer torque M in mind, which have been introduced in Figure 2.4, 2.8 and 2.10.

3.1 Stability

Since rider-controlled stability is the main focus of this thesis, several fundamental definitions and theorems that are continuously used will be stated in this section.

As already indicated in Section 2.1, stabilization in this thesis will stand for the return of the bicycle to the equilibrium at $\phi_0 = \delta_0 = 0$, after a disturbance. Therefore, asymptotic stability is required, which is defined e.g. in [20]:

Definition 3.1 (Lyapunov stability). *The equilibrium $x = 0$ of the system $\dot{x}(t) = Ax(t)$ is stable if for each $\epsilon > 0$ there exists a $\delta > 0$, such that for all initial conditions that satisfy*

$$\|x(0)\| < \delta \quad (3.1)$$

the inequality

$$\|x(t)\| < \epsilon, \quad \forall t > 0 \quad (3.2)$$

holds. The equilibrium is said to be asymptotically stable if it is stable and

$$\lim_{t \rightarrow \infty} \|x(t)\| = 0 \quad (3.3)$$

is fulfilled.

A criterion to prove asymptotic stability is also given in [20]:

Theorem 3.1 (Stability criterion). *The equilibrium $x = 0$ of the system $\dot{x}(t) = Ax(t)$ is asymptotically stable if all real parts of the eigenvalues Λ_i of the matrix A fulfill the inequality*

$$\Re(\Lambda_i) < 0. \quad (3.4)$$

The calculation of these eigenvalues, which are the poles of the system, is often very extensive. Therefore, the Hurwitz criterion is continuously used to prove if all poles have negative real parts and are therefore in the open left-half-plane (LHP). This criterion stated in [12] and [20] follows as:

Theorem 3.2 (Hurwitz criterion). *The polynomial*

$$p(s) = a_n s^n + a_{n-1} s^{n-1} + \dots + a_0, \quad a_n > 0 \quad (3.5)$$

is Hurwitz if all roots s_i of $p(s)$ are in the open LHP. This is the case if all n “northwestern subdeterminants” of the $n \times n$ matrix

$$H = \begin{pmatrix} a_{n-1} & a_{n-3} & a_{n-5} & a_{n-7} & \cdots \\ a_n & a_{n-2} & a_{n-4} & a_{n-6} & \cdots \\ 0 & a_{n-1} & a_{n-3} & a_{n-5} & \cdots \\ 0 & a_n & a_{n-2} & a_{n-4} & \cdots \\ 0 & 0 & a_{n-1} & a_{n-3} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{pmatrix} \quad (3.6)$$

are positive.

Another central method to prove stability in this thesis is the root locus, which is also described in [20]:

Definition 3.2 (Root locus). *The root locus is the geometric location of the poles of the closed-loop in the complex plane. The root locus illustrates the location of the poles depending on the gain k_p .*

Further fundamental methods and assumptions like phase or gain margin can be found in [12] and [20].

3.2 Parameters of the Rear-Steered Bicycle

The RSB which will be considered is illustrated in Figure 1.1. For simulations the parameters for the different bicycle models have to be estimated or, if possible, should be measured or calculated. A possibility to determine several essential parameters will be presented in the following.

At first, to get the CG of the bicycle it has to be hanged up at two different points one after another. In both positions a band is applied in vertical direction at the attachment. Since the bicycle is in a static equilibrium, the CG of the bicycle has to be on the intersection of the two bands.

Furthermore, to get the horizontal position a_r of the CG of the rider on the bicycle, a scale is put under a wheel of the bicycle with a test person. With the information of the scale, the mass of the test person and the mass and position of the CG of the bicycle, a static

torque balance can be set up around the ground contact point of the wheel without the scale. Additionally, it is assumed that this a_r holds as well for a standard rider with an average mass m_r given in [2].

The horizontal distance a of the CG of the system to the front-wheel can now be calculated with the principle of linear momentum using m_r and m_b . Finally, the height h_r of the CG of the rider sitting on the bicycle has been estimated and used for an analog calculation of the height h of the CG of the system.

These and further parameters can be found in Table 3.1. Moreover, this table only provides a brief description since more details are presented in the “List of Symbols and Abbreviations” or in the appropriate sections of this thesis.

Besides, it is important to note that for a FSB the parameters a for the horizontal position represent the distance to the rear-wheel instead of the front-wheel for a RSB. Therefore, they have to be subtracted from the wheelbase l . The other parameters hold for RSBs and FSBs.

3.3 Basic Model

Since the Basic Model covers the most important effects, it is very suitable for fundamental analyses. Consequently, it will be the core of this thesis. At first, the model is transferred to the frequency domain with the Laplace transform, which yields

$$G_{\phi\delta}(s) = \frac{av - s + \frac{v}{a}}{hl s^2 - \frac{g}{h}} \quad (3.7)$$

for Equation (2.23) of the Basic Model and represents the transfer function from the steer angle δ to the lean angle ϕ . $G_{\phi\delta}$ has a right-half-plane (RHP) zero $z = v/a$ and two poles $p_{1/2} = \pm\sqrt{g/h}$. Here, it is important to note, that the zero can be easily influenced by varying the velocity v and the distance a . Additionally, the upright position of the uncontrolled Basic Model is unstable since one pole is always positive. Thus, self-stability cannot be reached even for a FWS, what is contradictory to a real FSB. However, self-stabilization is utile but not necessary, since the goal of this thesis is to balance

Table 3.1: Parameters for the RSB of the IST.

Parameter	Value	Description
λ	71°	head angle
μ	$0.5 \frac{\text{kgm}^2}{\text{s}}$	friction term
a	0.62 m	horizontal position of the system's CG
a_b	0.46 m	horizontal position of the bicycle's CG
a_r	0.64 m	horizontal position of the rider's CG
a_{sr}	0.42 m	a with a standing rider
g	$9.81 \frac{\text{m}}{\text{s}^2}$	gravitational acceleration
h	1 m	height of the CG of the system
h_b	0.52 m	height of the CG of the bicycle
h_r	1.08 m	height of the CG of the rider
h_{rb}	0.4425 m	lever arm to the CG of the upper body [2]
h_{sr}	1.2 m	h with a standing rider
l	0.985 m	wheelbase
m	83.9 kg	mass of the system
m_b	11.4 kg	mass of the bicycle
m_r	72.5 kg	mass of the rider [2]
m_{rb}	29 kg	mass of the upper body of the rider
m_s	2 kg	mass of the steering system
m_w	0.8 kg	mass of the controlled wheel
r_τ	0.08 m	castor offset
R_s	0.35 m	radius of the steering system
R_w	0.34 m	radius of the controlled wheel

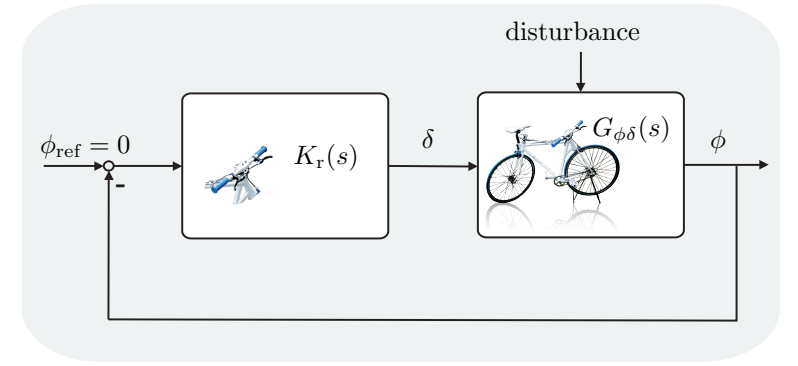


Figure 3.1: Closed-loop of the Basic Model of a RSB with a rider inducing a feedback.

a rider-controlled bicycle. Thus, the following sections will provide different rider models and controllers, to illustrate how stability can and how it cannot be achieved. Besides, a deeper insight into different effects discussed in Chapter 2 is delivered and fundamental recommendations are given to show what has to be done to make stabilization more easy.

3.3.1 Basic Rider Model

Since the bicycle has a RHP pole, a rider model is necessary to stabilize the RSB by providing a feedback. The closed-loop of the system is presented in Figure 3.1. Here, the rider $K_r(s)$ represents the controller and the bicycle $G_{\phi\delta}(s)$ represents the plant. Besides, the reference signal ϕ_{ref} has to be zero, since a stabilization in the upright position is aspired. A disturbance could be a wind gust or an unevenness of the road surface [20]. According to [6, 7], a simple model for the rider is

$$\delta = -k_\delta \phi. \quad (3.8)$$

This relation expresses that the rider steers into a lean with a certain gain k_δ , what corresponds to experience. Thus, the transfer function of the rider for this proportional controller is $K_r(s) = k_\delta$. Hence, the

transfer function of the closed-loop becomes

$$\frac{k_\delta G_{\phi\delta}}{1 + k_\delta G_{\phi\delta}}. \quad (3.9)$$

To provide stability, the denominator $1 + k_\delta G_{\phi\delta}$ has to be Hurwitz. After inserting the parameters, it can be found that

$$s^2 - k_\delta \frac{av}{hl} s + k_\delta \frac{v^2}{hl} - \frac{g}{h} \quad (3.10)$$

is not Hurwitz for any k_δ and consequently, the basic rider model cannot stabilize a RSB. In contrast to this, the FSB of Equation (2.24) can be stabilized with $k_\delta > \frac{lg}{v^2}$ [6, 7]. Accordingly, a FSB can be ridden with a very simple control law which makes it even easier to ride with higher velocities. This can be easily proven on a typical bicycle. Thus, to stabilize a RSB a more complex controller compared to a FSB is necessary, which makes riding a bicycle with RWS so difficult.

The reason for the difficult stabilization of a RSB is the non-minimum phase plant $G_{\phi\delta}(s)$, which causes an inverse response behavior due to the different direction of the kink force F_k . This non-minimum phase characteristic describes the relation between the centrifugal force F_{cf} and the kink force, both caused by the rider-controlled steer angle δ and $\dot{\delta}$ respectively.

Furthermore, a simulation of a bicycle with FWS and RWS with a velocity of $v = 4 \frac{m}{s}$ and a basic rider model with $k_\delta = 2$, is illustrated in Figure 3.2. Here, two different disturbances induce a fall of a bicycle to the right. In the different plots, the lean angle and the centrifugal and kink force are shown for the balancing process of a FSB and a RSB. Additionally, it can be seen that the kink force on a RSB is positive in the beginning in opposition to a FSB. Hence, as explained in Section 2.1.2 the kink force initially supports the disturbance and increases the lean angle before the steer angle becomes large enough to arouse a centrifugal force that can counteract this effect. This inverse response behavior has been simulated in the plot on the bottom left, at which the effect of the kink and the centrifugal force acts in the beginning in the direction of the disturbance. Compared to this, the FSB counteracts the disturbance from the start.

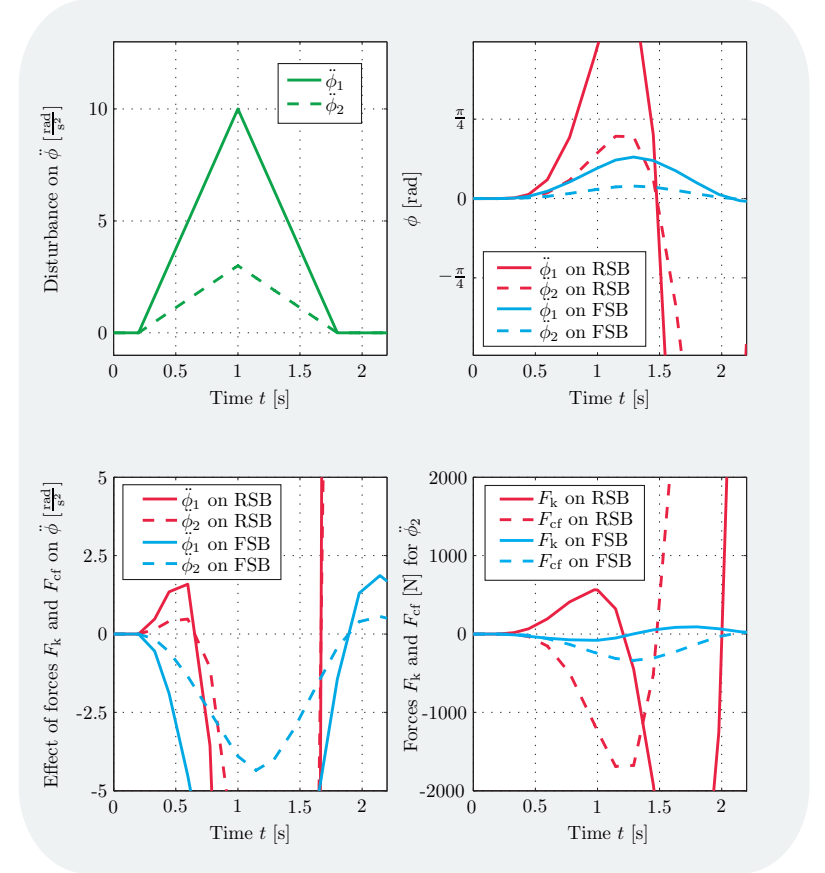


Figure 3.2: Behavior of a rider-controlled RSB and FSB at different disturbances with lean-steer.

Thus, with a kink force and a centrifugal force that act in the same direction (compare Table 2.1 and Figure 3.2), a FSB can be easily balanced with an appropriate lean-steer.

Moreover, for the different disturbances in Figure 3.2 there are two important scenarios for a RSB: In the first case there is a relatively large and fast disturbance $\ddot{\phi}_1$. Consequently, the rider will provide a large and fast steer angle and steer rate in the direction of the lean. As a result, the kink force further increases the lean angle ϕ which triggers that the rider steers even more in this direction. Finally, the centrifugal force is not able to counteract the lean early enough and therefore the RSB falls down to the right for $\phi > \frac{\pi}{2}$ [20].

In the second case the disturbance $\ddot{\phi}_2$ is slower and rather small. Thus, the steer rate of the rider and as a result the kink force will also be small. Consequently, the centrifugal force is large enough to recover the RSB at the beginning and causes a decrease of the lean angle at $t \approx 1.2$ s. However, the problem is that with this decrease, the kink force *and* the centrifugal force accelerate the bicycle in the direction of the upright position. Therefore, the kink force will be too large when the lean angle ϕ is zero, which can be seen in the plot on the bottom right in Figure 3.2. This makes it impossible for the centrifugal force to counteract the lean rate before the RSB falls to the left.

Besides, if a reverse lean-steer with $k_\delta = -2$ would be applied, the plots in Figure 3.3 can be obtained for $v = 4 \frac{\text{m}}{\text{s}}$. This illustrates that a reverse lean-steer is not a solution to the problem either. In fact, a reverse lean-steer makes riding a RSB even more difficult. The inverse response behavior of the RSB is illustrated in the plot of the “Effect of F_k and F_{cf} on $\ddot{\phi}$ ” on the bottom left. Here, the kink force F_k works against the disturbance, but after $t \approx 0.7$ s it is dominated by the centrifugal force F_{cf} which causes an overturn. The FSB is not rideable with a reverse lean-steer either, which can also be followed from Equation (3.10). Actually, the FSB falls faster than the RSB because the kink force *and* the centrifugal force act in the wrong direction in contrast to the RSB, at which only the centrifugal force causes a fall.

Finally, applying a reverse lean-steer is not useful for a RSB or a FSB and consequently further investigations will mainly concentrate

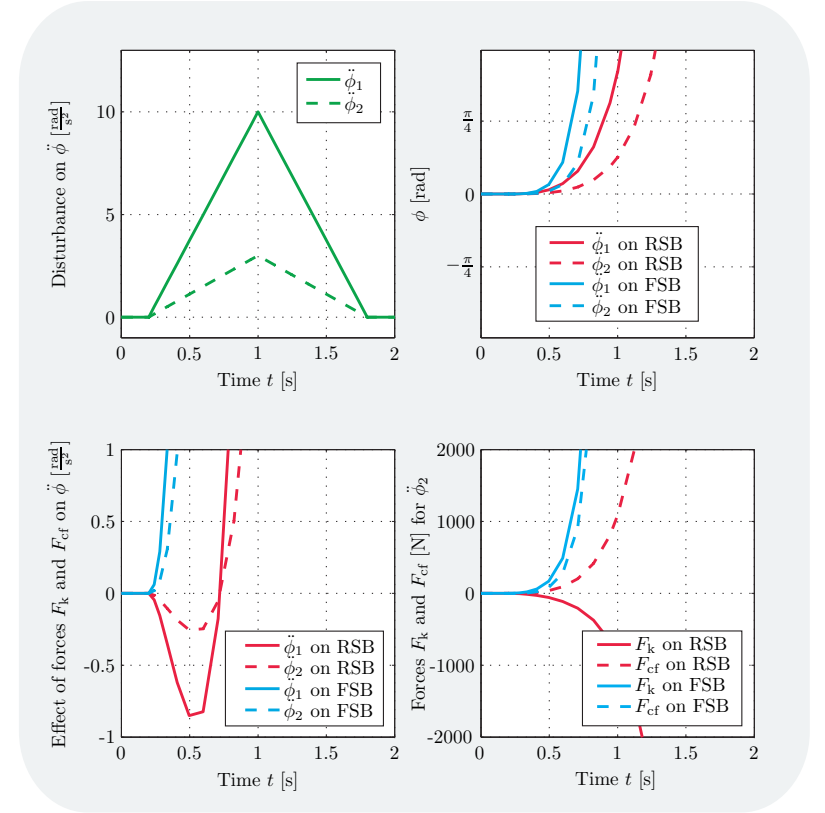


Figure 3.3: Behavior of a rider-controlled RSB and FSB at different disturbances with reverse lean-steer.

on steering into a lean. Additionally, to make the balancing process of a RSB realizable by a human rider, the following sections will often include a basic rider model as presented in this section. This basic rider model will then be the basis for a more complex controller design, which is indispensable for stabilizing a RSB.

3.3.2 Fundamental Limitations

Since the plant of the RSB has a RHP zero and a RHP pole, there occur fundamental limitations for the purpose of control caused by the dynamics of the system. This section will show that not just a more complex controller is necessary to stabilize a RSB but also an adaption of the plant. As the problematic RHP zero cannot be simply eliminated in the considered SISO feedback system (compare Section 3.3.3), the concern of the following is to optimize the preconditions and robustness for balancing a RSB.

The limitations based on the combination of a RHP zero and a RHP pole are stated in [3], which is the initial point of the following analysis. This scientific paper provides an equation to determine the optimal stability margins that can be reached with a certain configuration and gives recommendations to increase these margins. This equation and the approach to get there will be discussed in the following.

According to [3], the transfer function of the RSB model can be split up in a minimum and a non-minimum phase part G_{nmp} . The minimum phase part is comparatively easy to control, because there are no limitations due to system dynamics. Theoretically, this minimum phase part can be changed in a way that an arbitrary transfer function can be achieved [3, 4]. Thus, we assume to obtain Bode's ideal loop transfer function, which is

$$B(s) = \left(\frac{s}{\omega_{\text{gc}}}\right)^n. \quad (3.11)$$

Here, ω_{gc} is the crossover frequency of the gain, also called the bandwidth. Furthermore, the slope n is approximated as $n = n_{\text{gc}} = -\frac{1}{2}$, which is the slope at ω_{gc} . For a phase margin greater than or equal

to the optimal phase margin ϕ_m the inequality

$$\phi_m \leq \arg(G_{\text{nmp}}(j\omega_{\text{gc}})) + \frac{3}{4}\pi \quad (3.12)$$

can be stated [3], which limits the phase lag of G_{nmp} at ω_{gc} . Additionally, with the assumptions that $|G_{\text{nmp}}(0)| = 1$ and that G_{nmp} is stable in a closed-loop, the transfer function of the non-minimum phase part can be written as

$$G_{\text{nmp}}(s) = \frac{(z-s)(s+p)}{(z+s)(s-p)}. \quad (3.13)$$

Here, z is the RHP zero and p is the RHP pole of the RSB. Moreover, the crossover frequency is assumed to be located at the geometric mean $\omega_{\text{gc}} = \sqrt{pz}$ of the RHP pole and zero [3] to make an optimal stability margin possible. If the rider is not able to adjust the system's crossover frequency near to that optimal ω_{gc} , he cannot reach an adequate stability margin, since the phase lag of the non-minimum phase part will be too large (compare [4], page 11).

With the prerequisite $z > p$ and the determined relations for Equation (3.12), the optimal phase margin similar to [3] is given by

$$\phi_m = \frac{3}{4}\pi - 2 \arctan \frac{\sqrt{4\frac{z}{p}}}{\frac{z}{p} - 1}. \quad (3.14)$$

This equation gives a general answer to the question what stability margin can be achieved for a system with a certain RHP zero and pole.

If $z < p$ holds,

$$\phi_m = -\frac{5}{4}\pi - 2 \arctan \frac{\sqrt{4\frac{z}{p}}}{\frac{z}{p} - 1} \quad (3.15)$$

can be obtained analogically. However, as the velocity of the RSB has to be very small for $z < p$, riding with this condition would not be easy even with a FSB. Such a system is difficult to control, because the effect of the input is slow relatively to the unstable pole. Hence, usually more complicated controllers with amongst others large gains

Table 3.2: Optimal phase margin ϕ_m as a function of the ratio of the RHP zero and the RHP pole of a RSB.

	$\phi_m = 30^\circ$	$\phi_m = 45^\circ$	$\phi_m = 60^\circ$
ratio z/p	4.1	5.8	8.7

are necessary, which makes the stabilization for a human rider unrealistic (compare Section 4.1). Because of this additional difficulty, it will be assumed in further investigations that the rider pushes himself off the ground to reach a velocity such that $z > p$.

Moreover, it can be followed from Equation (3.14) that the ratio z/p needs to be large to assure an adequate stability margin. This correlation is illustrated in Table 3.2. If the ratio between z and p is not large enough, the system cannot be stabilized robustly with *any* controller.

Therefore, an appropriate adaption of the plant is necessary as indicated before. With $z = v/a$ and $p = \sqrt{g/h}$ it follows that a large velocity v , a large height h and a small distance a is necessary. This can be compared to the statements in [6, 7], at which it is recommended to stand up, lean forward and reach a high velocity very fast.

An interpretation of this technique for a RSB is, that the problematic kink force is reduced relatively to the centrifugal force by increasing v and decreasing a . Additionally, the unstable pole is slowed down by increasing h , which slows the capsize mode of the bicycle down and the rider has more time to react. Since the Basic Model has the same capsize mode as an inverted pendulum, a fast unstable pole could be compared to balancing a very short pendulum [27]. At a recumbent bike, this faster unstable pole also makes the starts more difficult. Another method for slowing the dynamics of the unstable pole down besides increasing h is illustrated in [6]. By applying wide tires with the rocking radius R_r , the pole p changes to

$$p = \sqrt{\left(1 - \frac{R_r}{h}\right) \frac{g}{h}}. \quad (3.16)$$

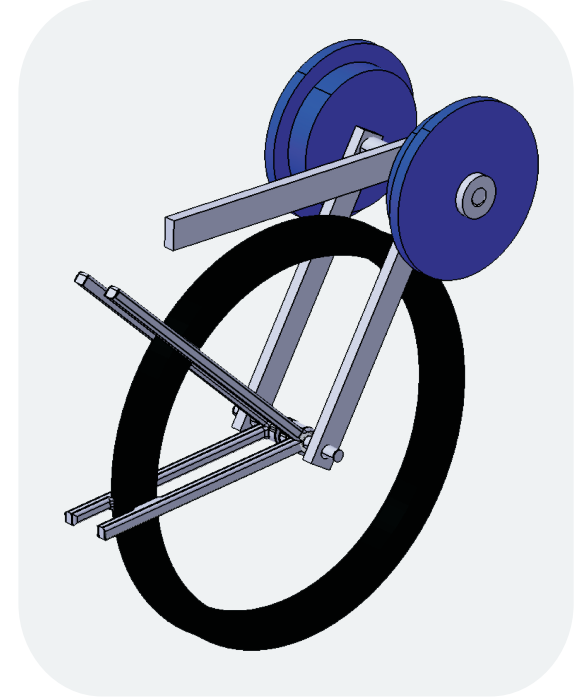


Figure 3.4: Three-dimensional sketch of a barbell applied to the front-wheel of a RSB to change the parameter a .

Thus, a large R_r could consequently slow down the unstable pole to provide an easier stabilization.

Furthermore, to make the zero faster the parameter a can be changed by leaning forwards or by applying e.g. a barbell to the front-wheel similar to Figure 3.4. Here, different weights can be used for varying a . Note that a change of a has a bigger effect on the ratio z/p compared to a change of h , since h is in the square root and a influences the ratio linearly.

Nevertheless, a high ratio between z and p only paves the way for an easier and robust stabilization. A more complex controller is still necessary, since a proportional controller cannot balance the RSB.

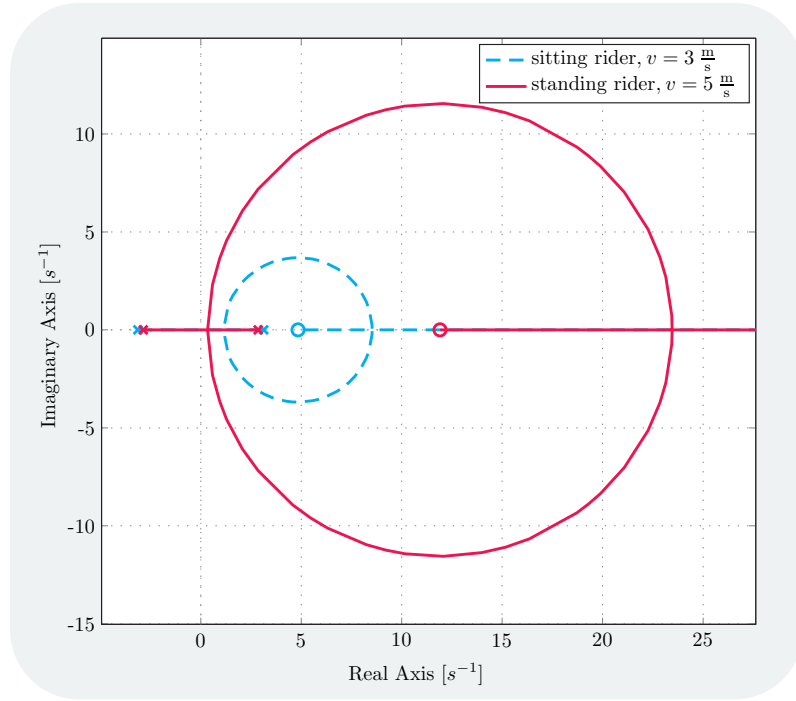


Figure 3.5: Root locus of a RSB with a sitting and with a standing rider at different velocities.

This can be seen in the root locus of Figure 3.5. Here, the blue curve uses the standard parameters $a = 0.62$ m and $h = 1$ m with a velocity of $3 \frac{\text{m}}{\text{s}}$. When the rider stands up with $h_{\text{sr}} = 1.2$ m, leans forward with $a_{\text{sr}} = 0.42$ m and rides faster with a velocity of $5 \frac{\text{m}}{\text{s}}$ the root locus gets closer to the LHP but still stays unstable, independent of the (positive) values of the parameters.

For this standing rider, the optimal phase margin ϕ_m is approximately 30° (compare Table 3.2). In contrast, the sitting rider would need a speed of $8 \frac{\text{m}}{\text{s}}$ to be able to stabilize the RSB with the same stability margins. The difference between sitting and standing has been tested with the RSB of the IST. As a result, this nearly un-

ridable RSB could be ridden at least for some meters when standing and leaning forward.

The relevance of adapting the plant has also been demonstrated in [15]. In this article, a bicycle is illustrated at which the rider is positioned pretty high over the front-wheel providing a large h and a very small a , which makes this bicycle indeed rideable. Finally, since the results of this section show what has to be done to be able to stabilize a RSB robustly, all further investigations that consider a SISO system will assume a standing rider.

3.3.3 Leaning Rider

So far, a basic assumption stated in Section 2.2.1 was that the rider does not lean relatively to the frame. Therefore, we always considered a SISO system at which the discussed limitations of Section 3.3.2 occur, caused by the RHP pole and zero. However, according to Åström [6] rider lean as an additional control input can make it possible to eliminate the RHP zero to stabilize the RSB. This approach will be investigated in the following.

Åström [6] finds that the problem with a RWS could be alleviated if the RHP zero would not be invariant and thus could be eliminated. As stated in [6, 23], the linear system

$$\dot{x} = Ax + bu \quad (3.17)$$

$$y = cx \quad (3.18)$$

with

$$x \in \mathbb{R}^n, \quad u \in \mathbb{R}^p, \quad y \in \mathbb{R}^q \quad (3.19)$$

has an invariant zero at s if the matrix

$$\begin{pmatrix} sI - A & b \\ c & 0 \end{pmatrix} \quad (3.20)$$

is losing rank. This is never the case if either b or c has rank n . A RHP zero can thus be eliminated in a feedback system if multiple-inputs or multiple-outputs are available. Consequently, the first method to

eliminate the RHP zero is to measure all state variables. In theory this works and e.g. a state feedback could be used for pole placement to stabilize the closed-loop as mentioned in [15]. Nevertheless, this is practically not realizable for a human controller who has to react precisely on the results of exact working tilt and yaw rate sensors. Moreover, an observer could not fulfill the requirements to eliminate the RHP zero.

The second and more convenient method to eliminate the RHP zero is to apply an additional input on the system. This could be e.g. rider lean. A differential equation of the Basic Model of a RSB with lateral rider lean similar to [6] is given by

$$J\ddot{\phi} - mgh\phi = -J_{rb}\ddot{\gamma} + m_{rb}gh_{rb}\gamma - \frac{mahv}{l}\dot{\delta} + \frac{mv^2h}{l}\delta. \quad (3.21)$$

Here, γ is the lean angle of the upper body of the rider relative to the bicycle's frame and J_{rb} is the moment of inertia of the upper body of the rider relative to its axis of rotation. Furthermore, m_{rb} and h_{rb} are the mass of the upper body and the distance of the CG of the upper body to its axis of rotation. Inserting

$$J = mh^2, \quad J_{rb} = m_{rb}h_{rb}^2 \quad (3.22)$$

and the control laws

$$\delta = -k_{\delta}\phi \quad (3.23)$$

and

$$\gamma = -k_{\gamma}\phi \quad (3.24)$$

in Equation (3.21), yields

$$\begin{aligned} (mh^2 - k_{\gamma}m_{rb}h_{rb}^2)\ddot{\phi} - \frac{k_{\delta}mahv}{l}\dot{\phi} \\ + \frac{k_{\gamma}m_{rb}gh_{rb}l + k_{\delta}mv^2h - mghl}{l}\phi = 0. \end{aligned} \quad (3.25)$$

The appropriate block diagram is illustrated in Figure 3.6. To obtain

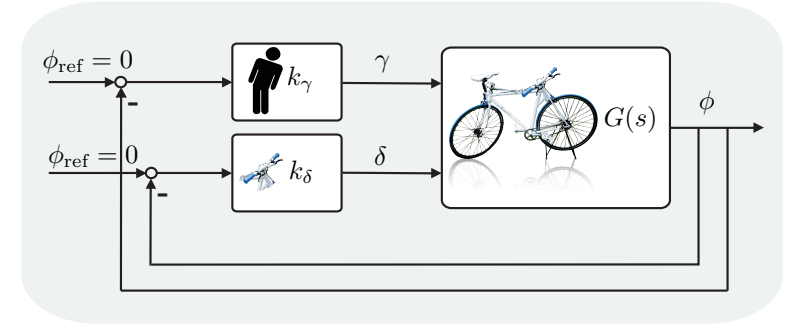


Figure 3.6: Closed-loop of the Basic Model of a RSB with a leaning and steering rider.

stability after the Hurwitz criterion for Equation (3.25), the three inequalities

$$mh^2 - k_{\gamma}m_{rb}h_{rb}^2 > 0, \quad (3.26)$$

$$-\frac{k_{\delta}mahv}{l} > 0 \quad \text{and} \quad (3.27)$$

$$\frac{k_{\gamma}m_{rb}gh_{rb}l + k_{\delta}mv^2h - mghl}{l} > 0 \quad (3.28)$$

have to be fulfilled. As a result of Equation (3.27), free-hand riding only with the use of rider lean is not possible, since k_{δ} can never be zero for asymptotic stability. This is as well the case for the Basic Model of a FSB. A plot of the resulting stability zone for different k_{δ} and k_{γ} over the velocity v is shown in Figure 3.7. Here, the standard parameters of Table 3.1 are used. Regarding the directions defined in Equation (3.23) and (3.24) it can be followed that the rider has to lean against the leaning direction of the bicycle and he has to apply a reverse lean-steer. Additionally, the velocity of the bicycle should be small. If the velocity becomes larger, k_{δ} has to be smaller or else the centrifugal force which is a function of v^2 will destabilize the RSB.

Besides, since this is no SISO system anymore and since the problem with the RHP zero can be solved, the results of Section 3.3.2 do not hold for a leaning rider. Thus, no general recommendation can be stated for robust stabilization.

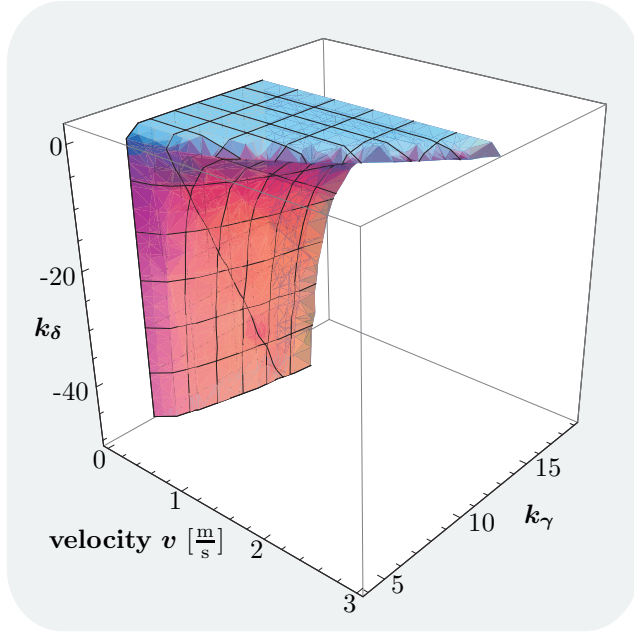


Figure 3.7: Plot of the stability zone of a RSB with different proportional controllers for steering and leaning as a function of the velocity.

To explain the technique of the rider to stabilize the system, a simulation for a disturbance $\ddot{\phi}_d$ applied on the bicycle with $k_\delta = -2$, $k_\gamma = 10$ and $v = 1 \frac{\text{m}}{\text{s}}$ is illustrated in Figure 3.8. Here, the bicycle begins to fall to the right. Then, the rider counters this by inducing the kink force F_k with a reverse lean-steer, which can be seen in the plot on the bottom right. Since the centrifugal force F_{cf} will grow with this reverse lean-steer and additionally supports the disturbance, a relatively large rider lean to the left is necessary to bring the bicycle back to the upright position by inducing the force F_r caused by the mass of the upper body of the rider.

In the plot on the bottom left, it can be seen that the forces applied through the rider do not show an inverse response behavior anymore. As a result, the RHP zero has been eliminated with rider lean as additional input and the RSB can be balanced, which can be seen in the plot on the top right.

However, it would be very difficult to realize this behavior. Because firstly, the stability zone is rather poor, secondly, the values k_γ for rider lean are very large as illustrated in Figure 3.7 and thirdly, no reaction time was assumed for an average value for upper body lean of about 0.3 s compared to applying a steer torque with about 0.1 s [6]. Hence, in the simulation illustrated in Figure 3.9 with a reaction time for upper body lean of 0.3 s, with $v = 1 \frac{\text{m}}{\text{s}}$, with $k_\gamma = 10$ and with $k_\delta = -2$, it can be found that the RSB is indeed not rideable if the rider does not react extremely fast.

Finally, since balancing with the considered technique is not relevant for a practical application, stabilization with the use of rider lean will not be further investigated. Nevertheless, there may be another way of stabilizing by leaning and steering, which could be the subject of future research. The focus for the following analyses and controller designs will be again the steer angle as “single-input”, at which the reaction time of the rider is much smaller.

3.3.4 Gyroscopic Effect

Since a more complex controller than a basic rider model is still necessary to stabilize the RSB, the gyroscopic effect of the spinning wheels

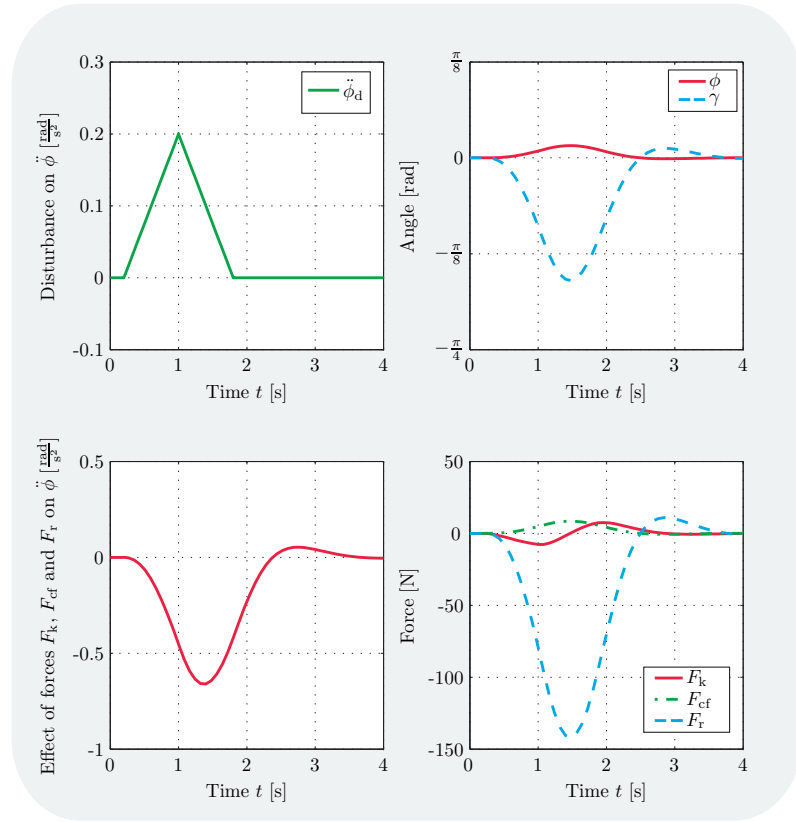


Figure 3.8: Behavior of a RSB for steering and leaning.

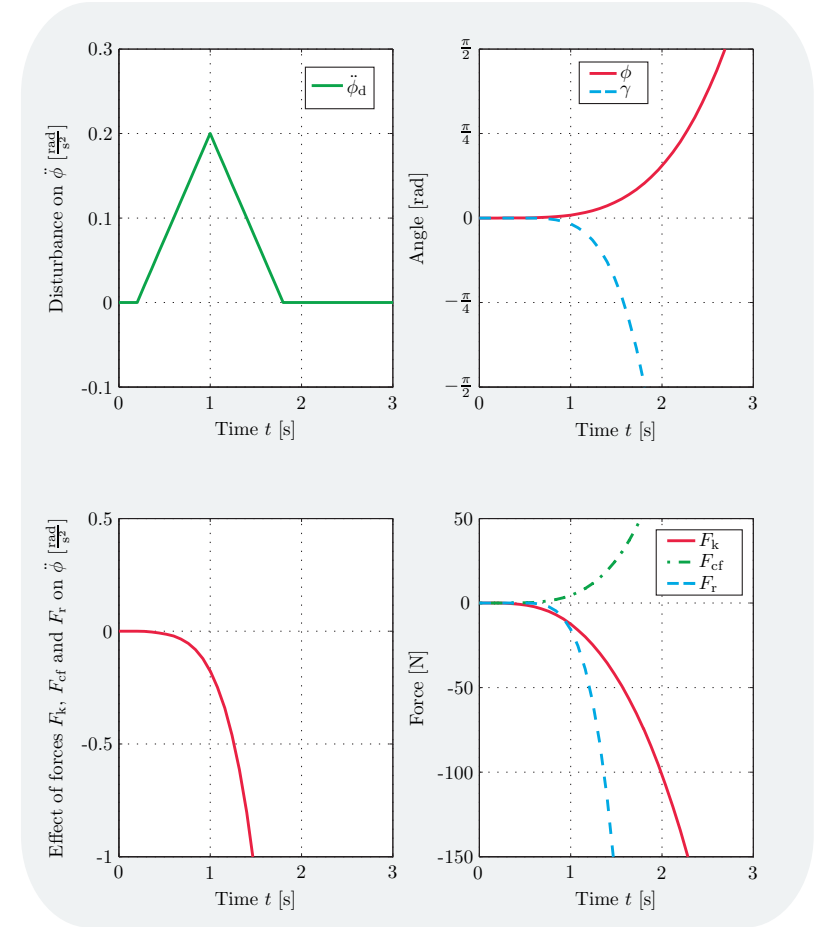


Figure 3.9: Behavior of a RSB for steering and leaning with a reaction time on rider lean.

which is illustrated in Section 2.1.4 will be taken into account. As it is often stated that this effect is important for bicycle stability, the following explanations will point out its fundamental implications on the Basic Model.

The torque M_{res} which represents the resulting gyroscopic effect on the controlled wheel, has been already introduced in Section 2.1.4. It is given as

$$M_{\text{res}} = \frac{dL}{dt} = |\vec{L}| \dot{\phi} = J_w \omega_w \dot{\phi}, \quad (3.29)$$

at which J_w denotes the moment of inertia and ω_w denotes the angular velocity of the controlled wheel [10]. Consequently, the lean rate $\dot{\phi}$ changes the direction of the angular momentum L which causes the resulting torque M_{res} .

Because the assumptions of the Basic Model contain no sliding and no loss of ground contact, the angular momentum is directly dependent on the velocity v , which changes Equation (3.29) to

$$M_{\text{res}} = \frac{J_w v}{R_w} \dot{\phi}. \quad (3.30)$$

Note that his final relation is similar to the gyroscopic effect implemented in the dynamic fork in Equation (2.42) for $\lambda = 90^\circ$.

Now, Equation (3.30) can be simplified and added to the basic rider model. According to Åström [5, 6]

$$\delta = -k_\delta \phi - k_g(v) \dot{\phi} \quad (3.31)$$

can be found for a FSB, at which k_g depends on the velocity. Since the gyroscopic effect will cause a steering into the opposite direction for a RSB (compare Section 2.1.4), the relation

$$\delta = -k_\delta \phi + k_g(v) \dot{\phi} \quad (3.32)$$

holds for a RWS. After a Laplace transform and a combination with the Basic Model of Equation (3.7), the denominator of the closed-loop can be expressed as

$$(1 + k_g \frac{av}{hl})s^2 - (k_\delta \frac{av}{hl} + k_g \frac{v^2}{hl})s + k_\delta \frac{v^2}{hl} - \frac{g}{h}. \quad (3.33)$$

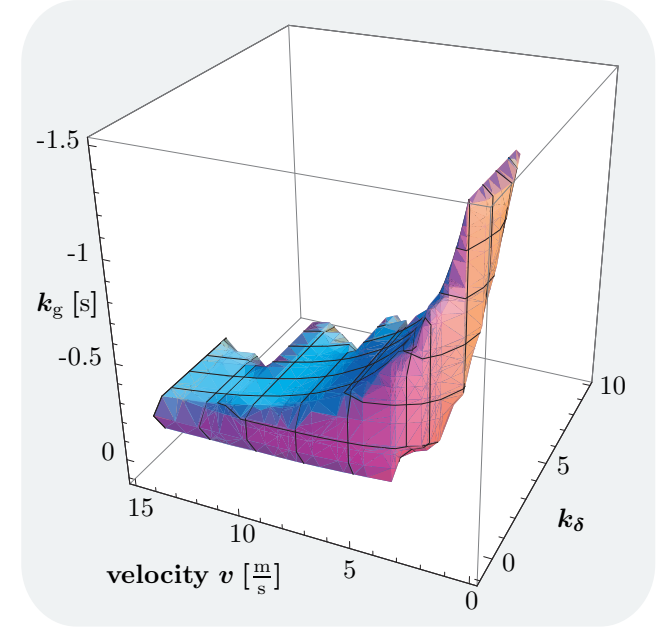


Figure 3.10: Plot of the stability zone of a RSB with gyroscopic effect and a standing rider.

From the Hurwitz criterion it follows, that k_δ needs to be positive and k_g negative for a stable configuration. Since k_g only changes the sign when the rear-wheel changes its direction of rotation, the RSB cannot be stabilized by scaling the gyroscopic effect.

However, if one would think of a “tricycle” there would be a possibility. Here, the third wheel should be hanging in the air and rotating backwards through a contact to the rear-wheel to change the sign of k_g . If the third wheel additionally has a much larger moment of inertia or a much smaller radius than the rear-wheel, its angular momentum will dominate the angular momentum of the rear-wheel. In this case all three terms of Equation (3.33) can be positive. For the parameters of Table 3.1 for a standing rider, the plot in Figure 3.10 shows the stability zone for a RSB with the gyroscopic effect. This zone be-

comes even bigger when a is further reduced and h further enlarged, what can be followed from the constraints of the Hurwitz criterion for Equation (3.33). A reason for this behavior is that with a smaller a the problematic kink force reduces, but the gyroscopic effect not. Note that this is also conform to the results of Section 3.3.2.

Furthermore, the transfer function of Equation (3.32) from ϕ to δ has the structure of an ideal PD controller that is

$$C(s) = k_p(Ts + 1), \quad (3.34)$$

at which k_p is a proportional controller and T a time constant. As a result, the technique of balancing a RSB with the gyroscopic effect and a basic rider model corresponds to performing a phase lead from ϕ to δ , which can be followed from the transfer behavior of Equation (3.34). This yields amongst others, that the steer rate $\dot{\delta}$ changes its direction before the lean angle ϕ reaches its maximum and minimum respectively. Since the kink force will consequently act in the correct direction together with the centrifugal force, the RSB can be balanced.

Besides, a reason for the ideal PD behavior is that no time lag has been considered. In a more detailed scenario Equation (3.34) would be a *real* PD controller or a lead compensator.

Finally, the results of this section are based on several significant simplifications beside the ideal behavior of the controller. Firstly, the rider could unintentionally counter the gyroscopic effect. Secondly, since k_g depends on the velocity v a theoretical stabilization could only be realized in a very small region of the stability zone of Figure 3.10. Thus, it would be very important to prove these findings with experiments. However, as this will not be done in the course of this thesis, the gyroscopic effect will be neglected in the following investigations since it is small for typical wheels. Nevertheless, as the principle of implementing a phase lead is a very useful and relatively simple approach, this will be discussed in detail in Chapter 4.

3.4 Static Fork Model

Within this section the more complex Static Fork Model of Section 2.2.2, which is stated by Åström [6], will be considered. This model additionally takes the head angle λ and the castor offset r_τ into account. In the following, some fundamental characteristics of this model will be analyzed. Moreover, the aim of this section is to look for effects that have been neglected for the Basic Model, which help to stabilize an uncontrolled RSB and a RSB with a basic rider model. It will be also discussed to what extent the Static Fork Model is applicable for a RSB, since it was basically derived to investigate self-stabilization of FSBs.

3.4.1 Self-Stabilization

At first, the Static Fork Model will be studied in regard to any configurations that could provide self-stability for a RSB. This would imply that the system is stable *without* an input by the rider. In this case not the rider, but the fork itself causes a stabilizing feedback. If the Laplace transform is applied to Equation (2.41) of the Static Fork Model of a FSB with a positive castor offset, the transfer function $G_{\phi M}$ from the steer torque M to the lean angle ϕ can be gained. The result of the Hurwitz criterion for $G_{\phi M}$ is illustrated in [6] as

$$v > v_c = \sqrt{lg \cot \lambda} \quad (3.35)$$

and

$$lh > ar_\tau \tan \lambda. \quad (3.36)$$

Here, Inequality (3.36) is typically fulfilled. Furthermore, Inequality (3.35) defines the critical velocity v_c above which the front fork causes a lean-steer behavior so that the FSB is self-stable. This corresponds to experience, since riding a bicycle without using hands is more easy with higher velocities.

Besides, the lean-steer behavior can be seen in Equation (2.36), at which M is zero and δ is proportional to $-\phi$, which represents a steering into a lean.

If the Equation (2.39) of the Static Fork Model for a RSB is considered, the constraints of the Hurwitz criterion change to

$$v < v_c = \sqrt{lg \cot \lambda} \quad (3.37)$$

and

$$lh < ar_\tau \tan \lambda. \quad (3.38)$$

Therefore, if the negative castor offset r_τ and the head angle λ are increased independently by attaching e.g. a rod, both constraints can be fulfilled and the model of this RSB is self-stable. Instead of only increasing the lever arm of the ground contact forces at the rear-wheel with c and r_τ , it would be also possible to adjust the position of the CG of the RSB with additional masses. Thus, a reduction of h and an increase of a would enlarge the normal force $F_{Z,W,\text{rear}}$ at the rear-wheel to provide self-stability.

Besides, for the following physical interpretation of these methods to stabilize the RSB, it is important to keep the derivation of the Static Fork Model in Section 2.2.2 in mind. According to Equation (2.36) the rear fork causes a reverse lean-steer for velocities $v < v_c$ and $M = 0$, which is necessary to fulfill Inequality (3.37). If the RSB would fall e.g. to the left, the rear fork would induce a steering to the right which finally moves the rear-wheel to the left. With an a , a castor offset and a head angle that are big enough to fulfill Inequality (3.38), the lateral displacement of the ground contact point of the rear-wheel causes a torque that is able to counteract the gravity torque. With this and with a stabilizing kink force caused by a reverse lean-steer, the RSB is theoretically self-stable.

Besides, a positive castor offset does not provide self-stability, since the inequality

$$lh + ar_\tau \tan \lambda < 0 \quad (3.39)$$

cannot be fulfilled for the positive geometric parameters and a λ which is defined between 0° and 90° . The reason for this is, that the static rear fork neglects the change of direction of the lateral tire force at the steered wheel of a RSB from transition period to steady-state (compare Figure 2.1). Consequently, the lateral tire force which is

always in a steady-state, is continuously stabilizing the steered rear-wheel of a RSB with a negative castor offset and destabilizing it with a positive castor offset (compare Section 2.1.5).

In addition to this neglected change of direction of the lateral tire force, there are further questionable assumptions and characteristics on which the theoretical stabilization with a negative castor offset is based. Firstly, the model is not conform to reality for velocities $v < v_c$, since a fork does not change its behavior of lean-steer to reverse lean-steer with velocity. That a typical bicycle provides a lean-steer even at $v = 0$ can be tested by holding the saddle and leaning the bicycle to one side.

Secondly, as the rear fork is assumed to be static it is always in a steady-state and consequently the torque caused by the lateral displacement is induced infinitely fast, since it shows no time lag. Assuming this, Inequality (3.37) expresses that the RSB is even stable with $v = 0$, which is not corresponding to reality.

Finally, inasmuch this model seems to be not applicable for low speeds without a rider, it will be investigated in the following if there are useful results for a rider-controlled RSB.

3.4.2 Basic Rider Model

Now, a rider will be integrated in the Static Fork Model to analyze how the RSB could be stabilized by a human. For the rider-controlled bicycle the closed-loop can be stated analog to Figure 3.1, at which the plant is the combination of frame and fork appropriate to Equation (2.39) for the RSB and Equation (2.41) for the FSB. In the following, the input of the rider will be the steer torque M instead of the steer angle δ .

First of all, the FSB will be covered with a basic rider model

$$M = -k_M \phi \quad (3.40)$$

corresponding to [6], to analyze the usability of a static fork for a rider-controlled bicycle. With this feedback of the rider, the closed-loop leads to the constraints of the Hurwitz criterion that are

$$\frac{magvr_\tau \sin \lambda + k_M lv}{(v^2 \sin \lambda - lg \cos \lambda)mhr_\tau} > 0 \quad (3.41)$$

and

$$\frac{mar_{\tau}g^2(lh \cos \lambda - ar_{\tau} \sin \lambda) + k_M l(v^2 h - ar_{\tau} g)}{(v^2 \sin \lambda - lg \cos \lambda)mah^2 r_{\tau}} > 0. \quad (3.42)$$

To fulfill these inequalities, the rider-controlled FSB is obviously stable for $v > v_c$, since it will be even self-stable in that case. Moreover, a rider does not cause instability if the steer torque M does not change the lean-steer behavior of the fork into a reverse lean-steer. Therefore the term on the right-hand side of Equation (2.36) which is

$$\delta = \frac{l^2}{(v^2 \sin \lambda - lg \cos \lambda)mar_{\tau} \sin \lambda} M - \frac{lg}{(v^2 \sin \lambda - lg \cos \lambda)} \phi, \quad (3.43)$$

has to stay negative.

However, for velocities smaller than v_c the rider can only cause a lean-steer behavior by applying a reverse lean-steer, what is very contradictory. Since he cannot influence the sign of $v^2 \sin \lambda - lg \cos \lambda$ with a positive steer torque M , this model seems to be not consistent for $v < v_c$, even with a rider. Besides, a change from a positive to a negative castor offset does not change this inconsistency. Hence, further calculations and interpretations would not be reasonable for small velocities.

At small velocities the Basic Model shows a more accurate behavior, since the rider can balance a slow FSB by a larger lean-steer k_{δ} . This can be followed from the constraint $k_{\delta} > \frac{lg}{v^2}$, which was derived in Section 3.3.1 and correlates to what we typically experience when riding a FSB.

For a RSB with a basic rider model the constraints

$$\frac{magvr_{\tau} \sin \lambda + k_M lv}{(v^2 \sin \lambda - lg \cos \lambda)mhr_{\tau}} < 0 \quad (3.44)$$

and

$$\frac{mar_{\tau}g^2(lh \cos \lambda - ar_{\tau} \sin \lambda) + k_M l(v^2 h - ar_{\tau} g)}{(v^2 \sin \lambda - lg \cos \lambda)mah^2 r_{\tau}} > 0 \quad (3.45)$$

of the Hurwitz criterion need to be fulfilled. Analog to the results of the FSB, velocities with $v < v_c$ will not be considered since the equation of the static rear fork is the same as for the static front fork.

To stabilize a RSB for velocities $v > v_c$, the rider needs to induce a reverse lean-steer with $k_M < 0$, to fulfill Inequality (3.44). If the parameters of this inequality are compared to Equation (3.43) of the rear fork, it can be found that the rider has to apply a larger reverse lean-steer than the lean-steer caused by the fork. Thus, a stabilizing kink force is induced. Furthermore, to fulfill Inequality (3.45), the destabilizing torque caused by the centrifugal force should be decreased by an appropriate reduction of h or v . Similarly, an adequate adaption of the torque of the lateral displacement is necessary as well.

Besides, a change of the castor offset to a positive configuration does not deliver any stable configurations.

Finally, this principle of using a reverse lean-steer to trigger a stabilizing kink force and the lateral displacement is a very theoretical concept that can hardly be realized. Besides, it is still based on the same questionable assumptions and characteristics as indicated at the riderless RSB. These are amongst others, that the tire force at the steered wheel is always in a steady-state and that the rear fork is infinitely fast. Therefore, the Static Fork Model should as well not be used for a rider-controlled RSB. Moreover, as the extracted principle of using the kink force in combination with the lateral displacement to stabilize the RSB has no practical relevance, it will not be integrated in further investigations. Hence, the Static Fork Model should only be concerned for a FSB to show the effect of self-stabilization.

To sum up, a static fork implies several problems. Thus, a dynamic fork will be considered in the next section.

3.5 Dynamic Fork Model

In the previous section it has been illustrated, that the Static Fork Model should not be used for a RSB. Hence, the Dynamic Fork Model will be investigated for effects that may have been neglected for the

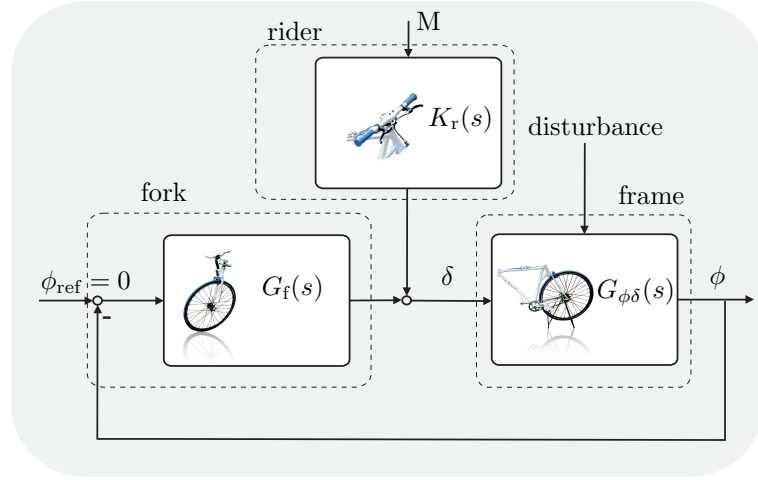


Figure 3.11: Closed-loop of the rider-controlled Dynamic Fork Model.

Basic Model, which can help to stabilize a RSB with a basic rider model.

3.5.1 Self-Stabilization

At first, the fourth order Dynamic Fork Model will be analyzed in regard to self-stability. Concerning this, Klein and Sommerfeld [14] have stated in the year of 1910, that a FSB is only self-stable in a certain velocity range. This behavior has been confirmed amongst others by [22]. With the Static Fork Model for the FSB it was not possible to point out this attribute. To verify, if the Dynamic Fork Model is detailed enough to show such a “self-stable region”, the model will be simulated in the following.

Therefore, the closed-loop of the dynamic fork and the frame has to be considered, which is illustrated in Figure 3.11. In the case of a riderless bicycle, the steer torque M is set to zero. Additionally, the transfer functions for a FSB and a RSB can be obtained by applying a Laplace transform on the appropriate equations of Chapter 2. That

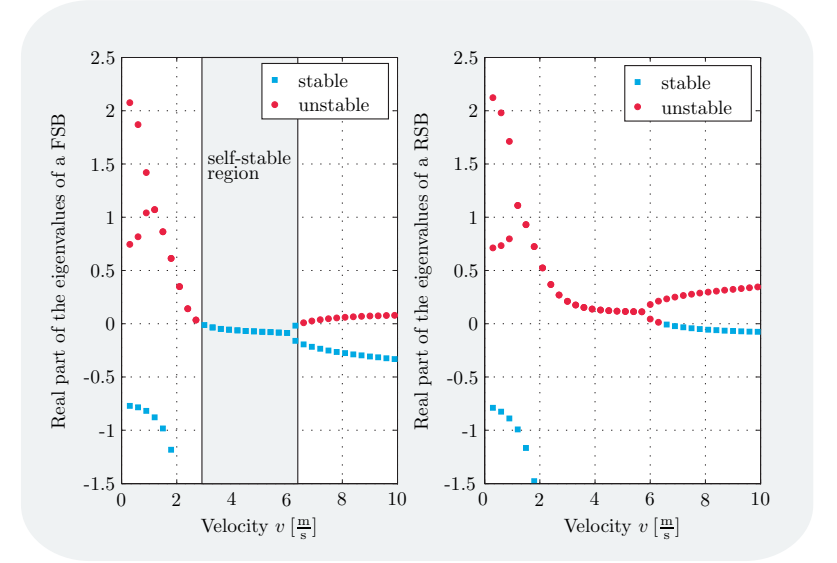


Figure 3.12: Plot of the eigenvalues of a riderless FSB and RSB as a function of the velocity.

is Equation (2.45) for the frame and Equation (2.44) for the dynamic fork of a RSB and analog to this, Equation (2.46) is used for the frame and Equation (2.47) for the dynamic fork of a FSB.

Building on that, it is useful in the case of a RSB and a FSB with standard parameters (compare Table 3.1), to illustrate the stability by a calculation of the eigenvalues as a function of the velocity. The results are shown in Figure 3.12. Note that in both plots the eigenvalues with real parts smaller than -1.5 are not shown, since the focus is the changeover from positive to negative.

From the plots it can be concluded, that the Dynamic Fork Model of a FSB features the mentioned self-stable region found by Klein and Sommerfeld [14]. Besides, it can be observed that the riderless RSB is not self-stable at any velocity for standard parameters.

Moreover, the Hurwitz criterion for a variation of the geometric parameters and the velocity v does not lead to a stable configuration

of a RSB either. As a result, the self-stable configuration of Section 3.4.1 for the Static Fork Model of a RSB cannot be confirmed with the Dynamic Fork Model. Furthermore, the bicycle simulation software JBIke6 [26] does not approve the results of the self-stable Static Fork Model either, which additionally supports the statement that the Static Fork Model is not adequate for a riderless RSB.

3.5.2 Basic Rider Model

Now, suppose a rider is controlling the RSB. Thus, the system can be investigated for techniques to stabilize the Dynamic Fork Model with a proportional controller. Here, the rider applies the steer torque M on the handlebar which can be seen in Figure 3.11. Hence, Equation (2.44) of the rear fork of the RSB changes to

$$\begin{aligned} & \frac{1}{4}m_s R_s^2 \ddot{\delta} + \mu \dot{\delta} \\ & + \left(-\frac{mgar_\tau \cos \lambda}{l} + \frac{mav^2 r_\tau \sin \lambda}{l^2} \left(1 + \frac{lm_w R_w \cos \lambda}{2ar_\tau m} \right) \right) \delta \quad (3.46) \\ & = \frac{m_w R_w v \sin \lambda}{2} \dot{\phi} - \frac{mgar_\tau \sin \lambda}{l} \phi + M. \end{aligned}$$

Note that the sign of the steer torque M in Equation (3.46) can be determined by a comparison to the direction of the gyroscopic effect on the rear fork, which is proportional to $\dot{\phi}$. Regarding this relation, the transfer function

$$\begin{aligned} & G_f(s) \\ & = \frac{-\frac{1}{2}m_w R_w v \sin \lambda s + \frac{1}{l}mgar_\tau \sin \lambda}{\frac{1}{4}m_s R_s^2 s^2 + \mu s + \left(-\frac{mgar_\tau \cos \lambda}{l} + \frac{mav^2 r_\tau \sin \lambda}{l^2} \left(1 + \frac{lm_w R_w \cos \lambda}{2ar_\tau m} \right) \right)} \quad (3.47) \end{aligned}$$

of Figure 3.11 can be expressed, as well as

$$\begin{aligned} & K_r(s) \\ & = \frac{1}{\frac{1}{4}m_s R_s^2 s^2 + \mu s + \left(-\frac{mgar_\tau \cos \lambda}{l} + \frac{mav^2 r_\tau \sin \lambda}{l^2} \left(1 + \frac{lm_w R_w \cos \lambda}{2ar_\tau m} \right) \right)}. \quad (3.48) \end{aligned}$$

With Equation (2.45) of the frame of a RSB, the transfer function

$$G_{\phi\delta}(s) = \frac{-ahv \sin \lambda s + (v^2 h - ar_\tau g) \sin \lambda}{lh^2(s^2 - \frac{g}{h})} \quad (3.49)$$

can be formulated.

Moreover, the rider will represent a proportional controller as already indicated. This yields the input

$$M = -k_M \phi. \quad (3.50)$$

However, utilizing the Hurwitz criterion for the closed-loop and varying the geometry, the velocity v and k_M does not result in any stable combination for a rider-controlled RSB. Consequently, there are no general recommendations that could be drawn from the Dynamic Fork Model with a basic rider model about additional stabilizing effects. This circumstance disproves as well the stable configuration of the Static Fork Model for a rider-controlled RSB.

Finally, since no new techniques for balancing a RSB could be extracted, only the Basic Model will be considered in the next chapter to stay focused on the most important effects. Here, further controller designs, which build on the results of Chapter 3 will be accomplished.

Chapter 4

Controller Design for a Rear-Steered Bicycle

In Chapter 3, some basic controllers have already been introduced to perform elementary investigations. In the following, they will be the basis for further more realistic controller designs that are able to stabilize a RSB. Moreover, they should still be realizable by a human rider.

The central assumptions for these controller designs are:

- The rider only influences the steer angle δ .
- The concept of phase lead of the gyroscopic effect in Section 3.3.4 will be the basis for the controller design.
- According to Section 3.3.2, a standing rider with $a_{sr} = 0.42$ m, $h_{sr} = 1.2$ m and a velocity of $v = 5 \frac{\text{m}}{\text{s}}$ will be assumed, so that a theoretical phase margin of $\phi_m \approx 30^\circ$ can be obtained.
- The Basic Model of the RSB will be considered.

4.1 Lead Compensator

The principle of phase lead of the gyroscopic effect of Section 3.3.4 naturally results in a lead compensator. Here, the rider has to implement the controller and therefore the phase lead himself.

To determine an appropriate and stabilizing lead compensator, a useful approach is discussed in [7]. Starting with the root locus of a proportional controller similar to Figure 3.5, the lead compensator has to “move” this root locus into the LHP. Here, the idea is to change the “symmetry” of the poles of the bicycle by “moving” the LHP pole further in the direction of the negative real axis. Thus, a LHP zero with about the same value as the LHP pole of the bicycle and a LHP pole that is much faster will be added. Consequently, the rider model can be assumed as a lead compensator

$$K_r(s) = \frac{k_\delta(\frac{s}{3} + 1)}{\frac{s}{10} + 1}. \quad (4.1)$$

As a result, the root locus of the Basic Model of the RSB with a lead compensator and a basic rider model can be seen in Figure 4.1.

Furthermore, the root locus becomes even more advantageous if the LHP pole of the lead compensator is faster and if the LHP zero would be nearer to the LHP pole of the RSB. Accordingly, the phase lead would further increase. However, realizing these points would be even more difficult for a human rider, since a very small time lag and a very exact steering would be necessary. Consequently, this will not be integrated in the following consideration.

Moreover, for a distortion applied on the rider-controlled RSB with $k_\delta = 0.6$, the resulting balancing process can be seen in Figure 4.2. Here, the phase lead from the lean angle $-\phi$ to the steer angle δ can be observed. Corresponding to Section 3.3.4, the crux of this technique to stabilize the RSB is that the rider changes the direction of the steer rate $\dot{\delta}$ before the bicycle changes the direction of its lean rate $\dot{\phi}$. For this reason, the centrifugal force and the kink force will both lift the bicycle upright. Consequently, the problem with the “wrong” direction of the kink force can be alleviated.

However, to realize the lead compensator there are several difficulties to deal with. Amongst others, the rider has to be very experienced to know when the forces are big enough to steer in the opposite direction, which needs a certain prediction. Additionally, the small range of values for k_δ to stabilize the RSB makes the balancing even more difficult.

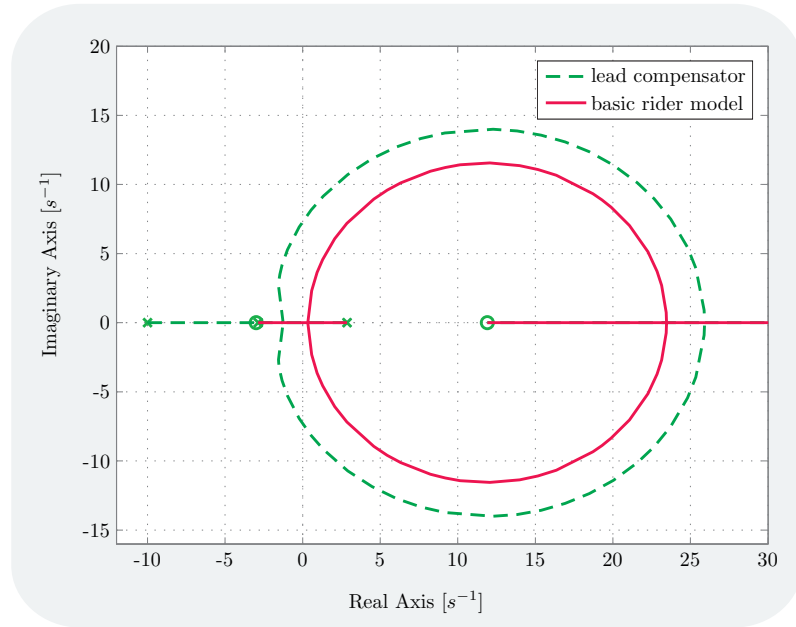


Figure 4.1: Root locus of the RSB with a standing rider representing a lead compensator and a basic rider model.

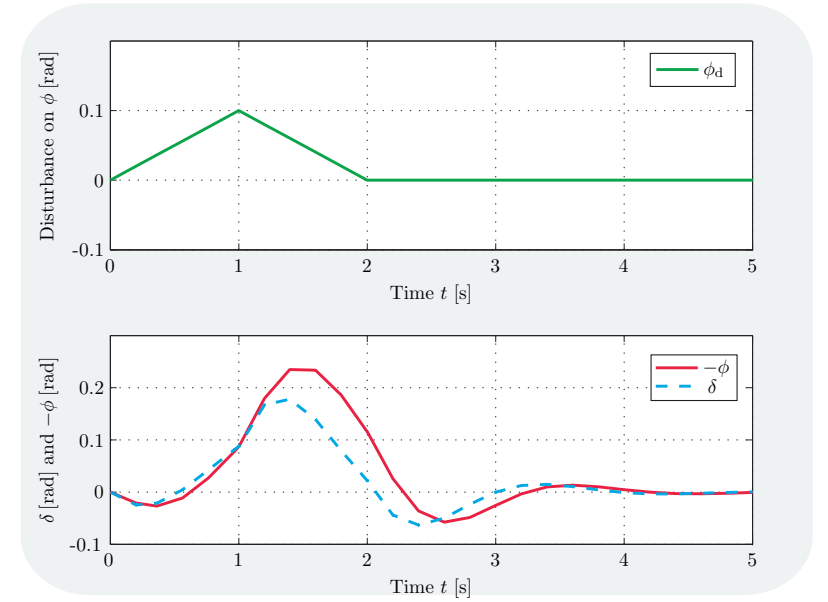


Figure 4.2: Relation between the steer angle and the lean angle, induced by a rider representing a lead compensator.

Moreover, if the lead compensator is written as differential equation, which is given as

$$\frac{k_\delta}{3}\dot{\phi} + k_\delta\phi = \frac{1}{10}\dot{\delta} + \delta, \quad (4.2)$$

it can be seen that the rider has to react not only to the lean angle ϕ but also to the lean rate $\dot{\phi}$. With this information he has to control not only the steer angle δ but also the steer rate $\dot{\delta}$, which demands a great deal of skill from the rider.

Besides, as indicated in Section 3.3.2, balancing a RSB is more complex if the RHP pole is larger than the RHP zero of the bicycle, which is the case for small velocities. Consequently, if the RSB should be stabilized with an easy concept, the rider should realize a transfer function similar to

$$K_r(s) = \frac{k_\delta(\frac{s}{3} + 1)}{-\frac{s}{10} + 1}. \quad (4.3)$$

To balance a RSB with a velocity of $v = 0.3 \frac{m}{s}$, the rider representing Equation (4.3) would need to provide a k_δ larger than 36. Since this is extremely difficult to perform and since this transfer function has an unstable pole, more complicated controllers have to be considered for such small velocities. Therefore, the reason for the assumption in Section 3.3.2 that the RHP zero is larger than the RHP pole, is accordingly the avoidance of the illustrated complexity. Hence, the rider should always ride fast enough for an easier and realizable stabilization.

However, as stated before realizing a lead compensator is not easy for a human rider even with larger velocities. For this reason, a concept will be introduced in the following section to support the rider.

4.2 Mechanical Controller

To facilitate the steering process of the rider for a RSB, a mechanical controller will be developed. The aim of this controller is, that a basic rider model combined with a certain mechanical construction can

stabilize the Basic Model of the RSB. Hence, a mechanical controller with a transfer function similar to a lead compensator is desired to provide a phase lead for the motion of the bicycle. Since a mechanical lead compensator is not realizable, the mechanical system with the transfer function

$$\frac{K_p(1 + T_{D1}s)}{1 + T_1s + T_2^2s^2 + \dots} \quad (4.4)$$

illustrated in [24] on page 83, would also provide the necessary phase lead that is able to keep the RSB upright with a basic rider model. Here, the parameters T_{D1} , T_1 and T_2 are time constants.

Another system presented in [24] on page 83 which is

$$\frac{K_p(1 + T_{D1}s + T_{D2}^2s^2)}{1 + T_1s + T_2^2s^2 + \dots}, \quad (4.5)$$

could also stabilize the RSB. But, since a mechanical realization would be much more extensive, only a configuration similar to (4.4) will be considered in the following.

As a SISO system is discussed, the position of the mechanical controller in the closed-loop is arbitrary. Thus, the configuration shown in Figure 4.3 is chosen, since it is least complex. Here, the mechanical controller changes the transmission of the steer angle from the handlebar to the rear-wheel. For other positions of the mechanical controller in the closed-loop, the construction would have to change the lean angle and provide a feedback which is very complicated to realize. However, at the position in Figure 4.3, the basic idea is that the controller implements an appropriate dynamic behavior to the transmission of the steer angle. Therefore, the rider output $\tilde{\delta}$ has to be modified by the transfer function $K_m(s)$ of the mechanical construction to achieve stability.

For the purpose of modeling the controller, a sketch of the steering mechanism with crossed wires of the RSB of the IST is illustrated in Figure 4.4. Here, the parameters c and k are spring constants, d is a damping coefficient and R_{cr} is the radius of the chain rings. Moreover, the angle $\tilde{\delta}$ is the input of the mechanical controller and denotes the rotation of the handlebar. And the angle δ represents

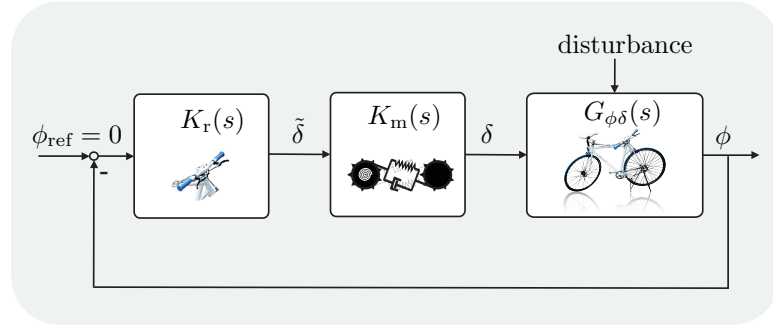


Figure 4.3: Closed-loop of the Basic Model of a RSB with a basic rider model and the mechanical controller influencing the steer angle.

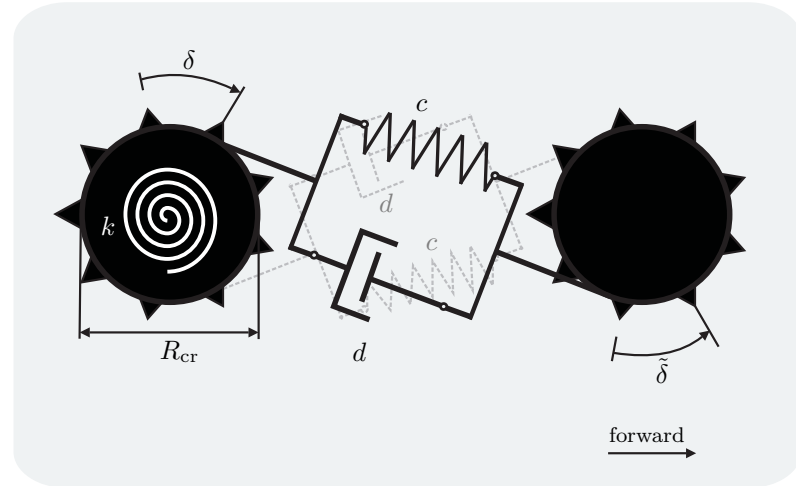


Figure 4.4: Top view of a sketch of the mechanical controller for a RSB.

the modified output and rotation of the rear-wheel. Note, that a part of the construction of the steering mechanism in Figure 4.4 is transparent to provide a better overview.

Since wires are used for the transmission, the steering mechanism can only transfer tensile forces. Therefore, only one side of the construction is active at a time and a “complete” model is equivalent to a model of only one side which is assumed to be able to transfer compressive forces as well. Subsequently, for such a model the principle of angular momentum can be written as

$$J_{\text{rear}}\ddot{\delta} = R_{\text{cr}}^2 d(\ddot{\tilde{\delta}} - \ddot{\delta}) + R_{\text{cr}}^2 c(\tilde{\delta} - \delta) - k\delta. \quad (4.6)$$

The estimated moment of inertia of the rear-wheel and the rear fork relative to the steer axis is given as $J_{\text{rear}} \approx \frac{1}{4}m_{\text{rear}}R_s^2$. Here, m_{rear} is the mass of the rear-wheel and the rear fork and R_s is the radius of a virtual cylinder containing this mass. Consequently, the transfer function of the mechanical controller follows as

$$K_m(s) = \frac{ds + c}{\frac{1}{4R_{\text{cr}}^2}m_{\text{rear}}R_s^2s^2 + ds + c + \frac{k}{R_{\text{cr}}^2}}. \quad (4.7)$$

Now, the closed-loop shown in Figure 4.3 has to be analyzed in regard to stability. To be able to choose appropriate values for c , d and k , the Hurwitz criterion will be applied. The necessary parameters for calculations are assumed as $m_{\text{rear}} = 1.4 \text{ kg}$, $R_{\text{cr}} = 0.03 \text{ m}$ and the basic rider model will be assumed as $K_r(s) = k_{\tilde{\delta}} = 1$ as this value is easy to realize.

In Figure 4.5 the stability zone is illustrated as a function of the components of the mechanical controller. Since the shape of this zone stays almost the same when varying the range of c , d and k , the ratio of these parameters is very important and wrong values could even worsen the problem. Considering Figure 4.5, the set of parameters with $c = 3000 \frac{\text{N}}{\text{m}}$, $d = 1250 \frac{\text{kg}}{\text{s}}$ and $k = 3 \frac{\text{Nm}}{\text{rad}}$ is chosen to provide realizable values. Here, the value for c is typical but k and d are rather large. Therefore, it would be important to conduct experiments to verify the theory.

Regarding the root locus of the system with the determined parameters in Figure 4.6, a certain similarity to the lead compensator

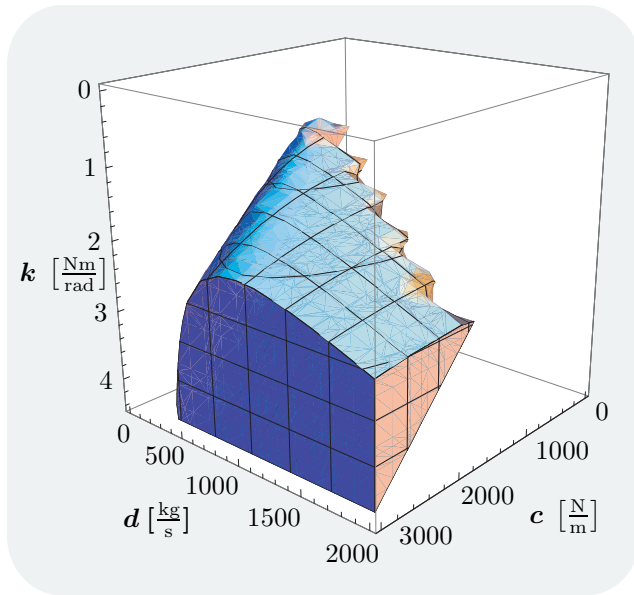


Figure 4.5: Plot of the stability zone of a RSB for different spring constants and damping coefficients of the mechanical controller.

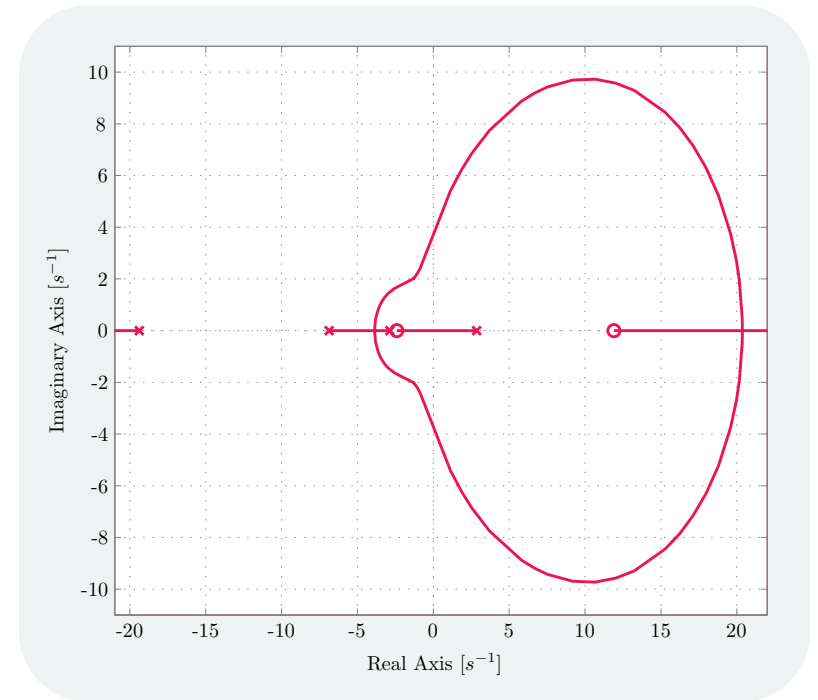


Figure 4.6: Root locus of a RSB with the mechanical controller and a basic rider model.

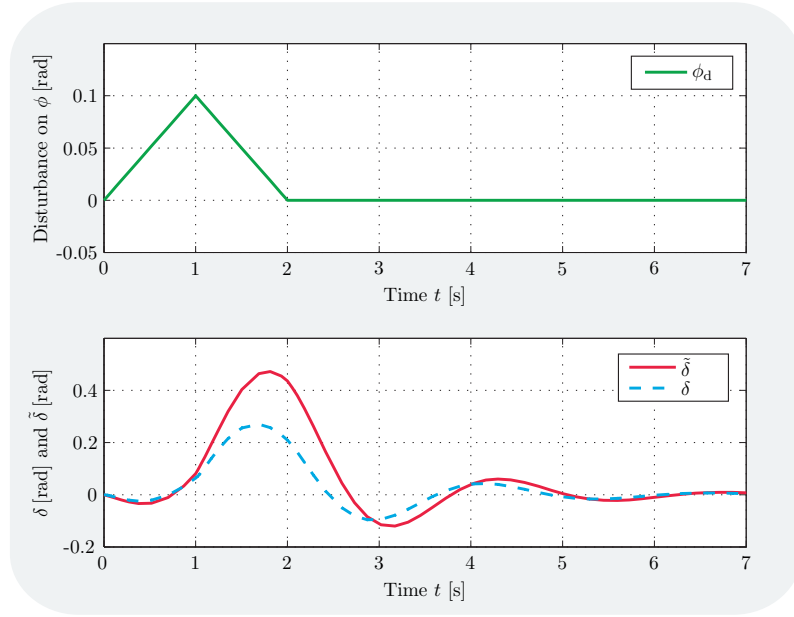


Figure 4.7: Relation between $\tilde{\delta}$ and δ induced by the mechanical controller.

can be found. In fact, the LHP zero of the mechanical controller is as well near to the LHP pole of the RSB and the LHP poles of the controller are much faster.

This similarity emerges as well in the simulation of a disturbance on ϕ of the system, which is shown in Figure 4.7. Like the lead compensator, the mechanical controller causes a phase lead from the lean angle $-\phi$, which is proportional to $\tilde{\delta}$, to the steer angle δ for the motion of the bicycle. As indicated in Section 3.3.4 and 4.1, this phase lead stabilizes the RSB. To analyze the reason for the illustrated behavior it is important to consider the mechanical components of the controller. Amongst others, one important effect is that the spring constant k is proportional to δ whereas c is proportional to the difference $\tilde{\delta} - \delta$ and d to the derivative of this difference. Therefore, the

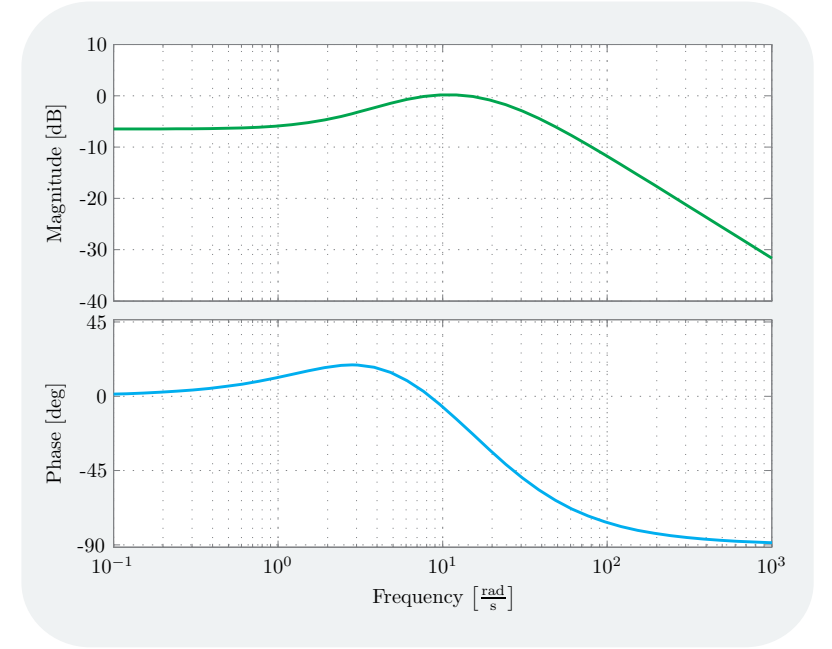


Figure 4.8: Bode plot of the mechanical controller.

term $-k\delta$ determines the sign of the right part of Equation (4.6) if the curve of $\tilde{\delta}$ and δ have a similar development. This causes $\dot{\delta}$ to decrease before $\dot{\tilde{\delta}}$ which finally slows the lean rate $\dot{\phi}$ down to provide stability.

Note that if the disturbance is much faster than in Figure 4.7, there will be a phase lag in the beginning, since the Bode plot of the mechanical controller illustrated in Figure 4.8 will have a negative phase for large frequencies. A reason for this behavior is, that the mass m_{rear} cannot be accelerated infinitely fast. Nevertheless, this phase lag will change to a phase lead when the disturbance declines. Because then, the RSB will turn with about its capsize mode of $\sqrt{g/h_{\text{sr}}} \approx 2.86 \frac{\text{rad}}{\text{s}}$ causing the mechanical controller to provide a positive and stabilizing phase.

In conclusion, the interaction of the determined mechanical components is theoretically able to stabilize a rider-controlled RSB. However, the stability margins are small. If the reaction time and the time lag of the rider would be considered, it will be very difficult to balance a RSB. Nevertheless, a human rider will gain experience when trying to ride a RSB, so that he will be able to represent a more complex controller using amongst others an appropriate rider lean to enhance the stability margins.

Chapter 5

Conclusion and Outlook

5.1 Conclusion

In the beginning of this thesis, the essential effects of bicycle dynamics have been introduced. With these effects three models of rear-steered bicycles have been derived by an adaption of the derivations of front-steered bicycles given by literature.

Afterwards, the equation of a Basic Model of a rear-steered bicycle was analyzed and simulated with a Basic Rider Model to find the reason for the inherent stability problems. As a result, the inverse response behavior and the restrictions caused by the dynamics of the non-minimum phase system of a rear-steered bicycle have been found to be the crux of the difficult stabilization. Further analyses, which led amongst others to a change of several parameters for robust stabilization, were mainly based on research by Åström [6]. Experiments with the rear-steered bicycle of the Institute for Systems Theory and Automatic Control showed that a variation of these parameters is enormously important to be able to stabilize the bicycle.

Based on these findings, several theoretical concepts have been developed in the course of this thesis, to show how a bicycle with rear-wheel steering can be stabilized. At first, it has been illustrated that the inverse response behavior, which causes the difficulties of riding a rear-steered bicycle, can be eliminated by additionally applying a rider lean as input beside the steer angle on the handlebar. But the

determined technique for these two inputs would not be practically applicable. Moreover, rider lean has a much larger reaction time than applying a steer angle. Thus, only models with a “single-input” on the handlebar have been considered afterwards.

Subsequently, the concept of the gyroscopic effect has been investigated. Here, the principle of phase lead could be extracted, which is central to balance a rear-steered bicycle only by steering. However, the gyroscopic effect implied significant simplifications. Furthermore, no experiments have been done to test these results. Thus, only the principle of phase lead has been used for further research. Moreover, based on the results of the gyroscopic effect, two controllers have been designed to stabilize the bicycle with rear-wheel steering. The first one is a lead compensator, which should be realized by a human rider. Nevertheless, since this would not be easy, a second and mechanical controller has been developed to support the rider to perform the necessary phase lead. For this mechanical controller, calculations showed that it is important to choose the appropriate parameters for the spring constants and the damping coefficient. Therefore, simplifications in the model could also cause problems using this technique.

Further analyses showed that a Static Fork Model should not be used for a rear-steered bicycle. Moreover, a Dynamic Fork Model did not show any new regions in design space for a stabilization with a basic rider model. Possibly, more detailed analyses and more complex controller designs for this bicycle model would lead to further improvements. However, since the Dynamic Fork Model was not of central interest in this thesis, this has not been investigated in detail.

The main result of this thesis is, that a rear-steered bicycle can be balanced by only using the steer angle as input and by applying a phase lead from the lean to the steer angle. The research done in the course of this thesis also showed that mechanical changes on the bicycle, e.g. a third wheel to influence the gyroscopic effect or a mechanical controller at the transmission of the steer angle, are necessary to theoretically make the stabilization easier for a human rider. Consequently, this thesis extended previous research, by describing in detail what could be done about the rear-steered bicycle and how a rider has to behave to stabilize it.

Furthermore, stabilization by only using the steer angle as input re-

sults in rather small stability zones, which are becoming even smaller when considering time lag and reaction time. These characteristics make the investigations more realistic, but since the rider represents a much more complex controller in reality, the use of e.g. appropriate rider lean would further increase the stability zones. Moreover, most of the results that improve stability can be combined to additionally support the balancing process.

However, there is obviously no way to make a rear-steered bicycle as easily rideable as a front-steered bicycle, since the dynamics are completely different.

5.2 Outlook

In this thesis, numerous concepts and techniques to make a rear-steered bicycle rideable have been illustrated. Therefore, it would be important to verify the presented theoretical results with experiments, to show their relevance for the balancing process and to combine several concepts to obtain the best results.

Moreover, fundamental analyses and controller designs have mostly been done for the Basic Model. Therefore, more detailed analyses could be performed for the Dynamic Fork Model implying nonlinear equations, further controller designs for rider lean or a variable velocity at the front-wheel.

Besides, a topic of interest, especially for the bicycle of the Institute for Systems Theory and Automatic Control, is a further analysis of the steering transmission. Using a cam follower for a variable input-output ratio as described by Cornelius [8], could e.g. support the balancing process. Furthermore, a change of the input-output ratio to get a decrease of the output, made riding the Institute's RSB easier. The reason for this effect is, that the chain drive which is used at the Institute's bicycle, has a relatively large backlash. Consequently, an appropriate change of the input-output ratio reduces the influence of the backlash, which makes it easier to steer more precisely. Based on this, the idea came up to change the chain drive to a stiff linkage that shows almost no time lag or backlash. Such a linkage has been used by David Bothman, who could indeed stabilize his rear-steered bicycle when riding fast enough [6].

Further research on rear-wheel steering could also deal with rear-steered recumbent bicycles. Cornelius [8] demonstrated, that they are much easier to ride than common rear-steered bicycles. Therefore, it would be interesting to find the reasons for this matter of fact in regard to a transfer of the results to a typical rear-steered bicycle.

Finally, the field of bicycle dynamics stays interesting. This year, scientists of the Delft University of Technology have built a two-mass-skate which is self-stable without gyroscopic effects and without any castor offset, what has been denied for a long time by many experts. Hence, it could be possible that there are still new and useful regions in design space to further improve bicycles [16].

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